ME - 780

Introduction to Vehicle Dynamics, Control, Hybrid/Electric Vehicles, and Automated Driving

Lecture 4: Longitudinal and Single-track Linear Handling Models

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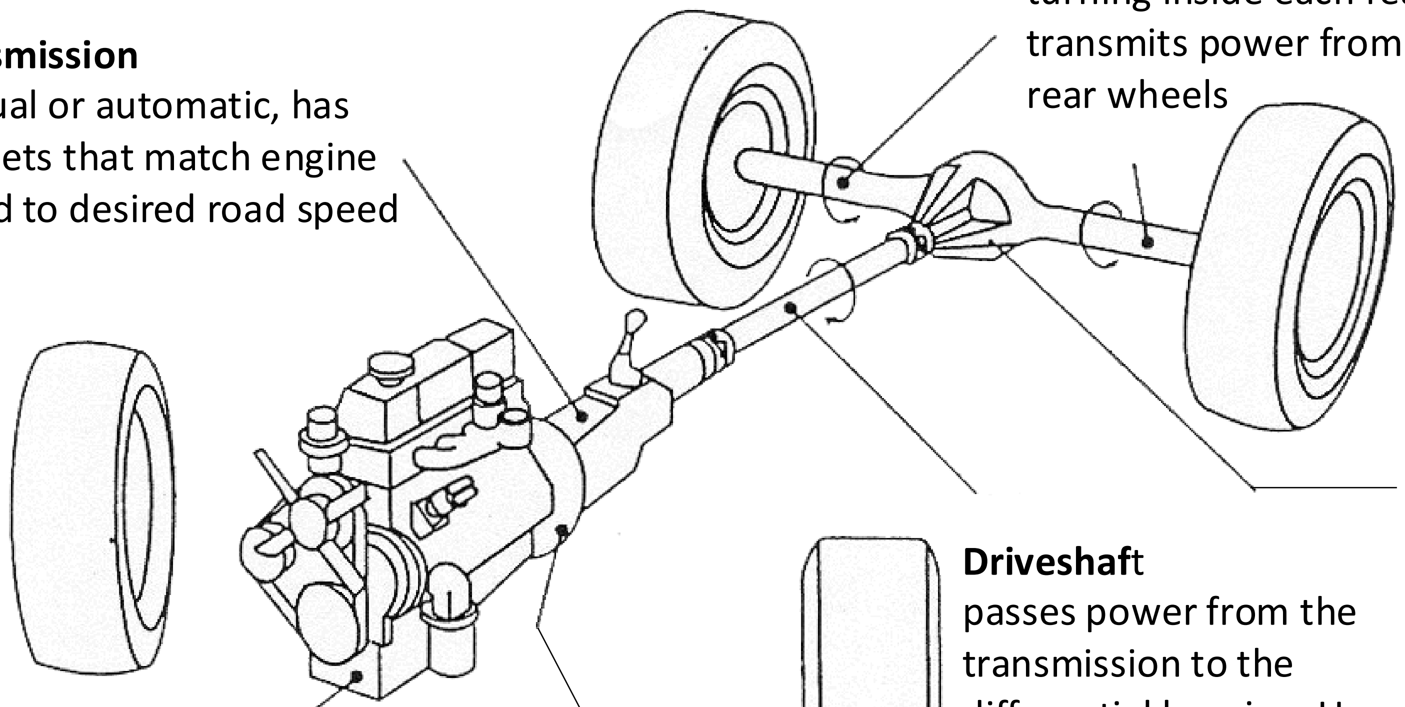
Contents

Longitudinal Vehicle Dynamics Model

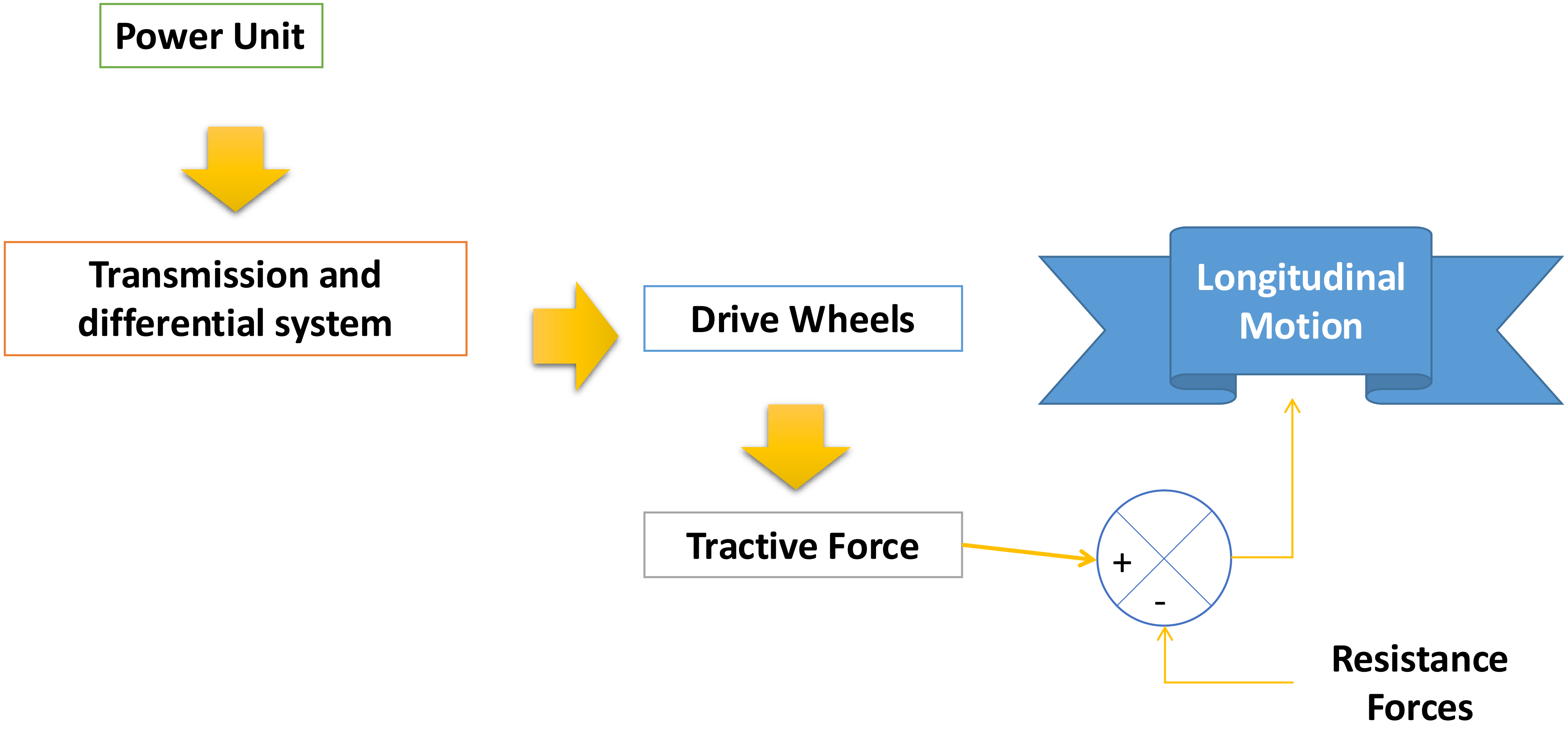
- **A vehicle model to perform vehicle performance including:**
  - **Top speed (in different road slopes)**
  - **Gradeability**
  - **Acceleration time**
  - **Fuel consumption**
- **Assumptions:**
  - **The vehicle accelerates or brakes during a straight-line driving,**
  - **Longitudinal velocity is the most effective degree-of-freedom (DOF),**
  - **Lateral motion as well as yaw, roll, and vertical motions are ignored,**
  - **Pitch motion has some second-order effects on the longitudinal load transfer,**
  - **The road is dry with high coefficient of friction, $\mu = 1$,**
  - **No lateral slip and small longitudinal slip (less than %5).**

Accelerating Model: Powertrain Configuration



The main elements of power train system

Accelerating Model: Vehicle Power Follow



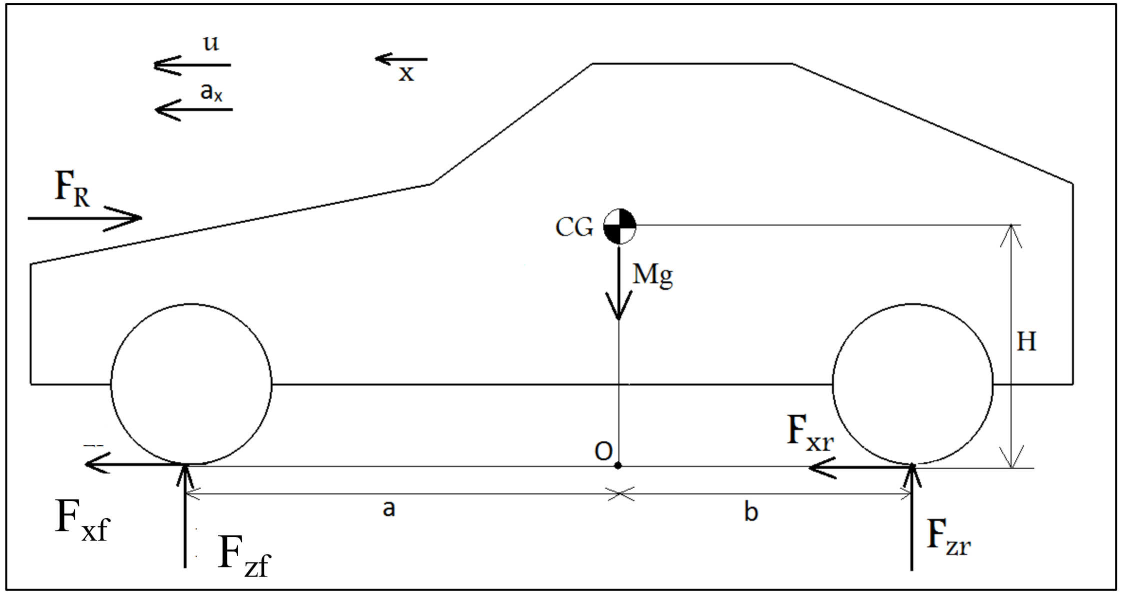
Accelerating Model: Longitudinal Equations of Motion

$$\sum F_x = M(\dot{u} - \overbrace{vr}^{\mathbf{0}})$$

$$(\underbrace{F_{xf} + F_{xr}}) - F_R = M\frac{du}{dt}$$

$$F_x = F_{Tractive}$$

Generated by powertrain



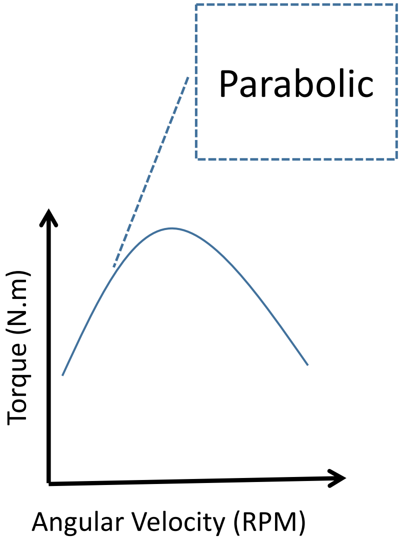
Accelerating Model: Power Unit Model

Power Unit

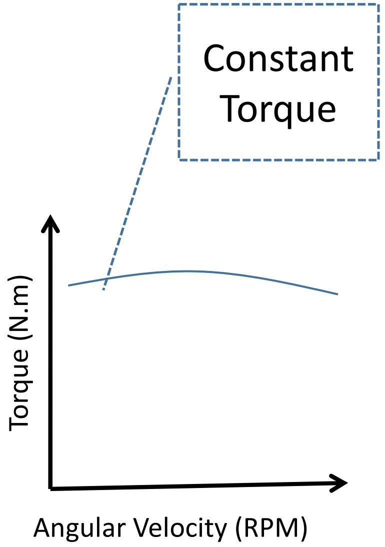
$X_\theta \rightarrow$ **Power Unit (Engine)** $\rightarrow \omega_e\ , T_e$

X_θ = Accelerator (gas) Pedal position

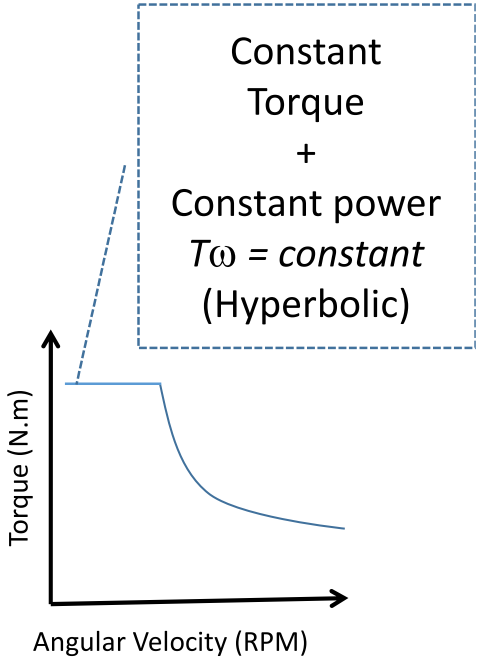
Different types of power units T-ω Curves



Petrol Engines



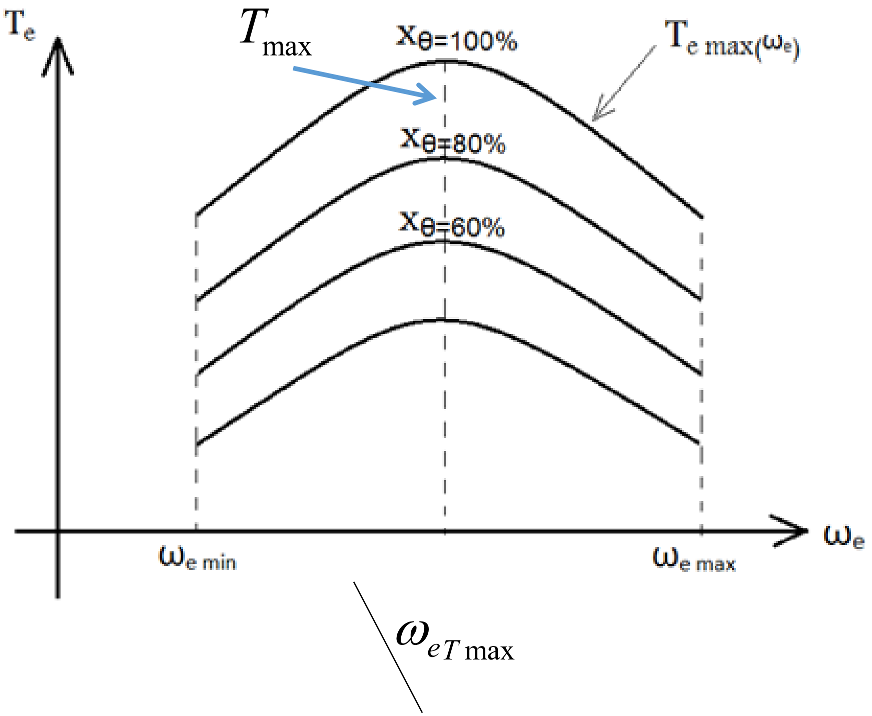
Diesel Engines



Electric Motors

Accelerating Model: Torque and Power Curves

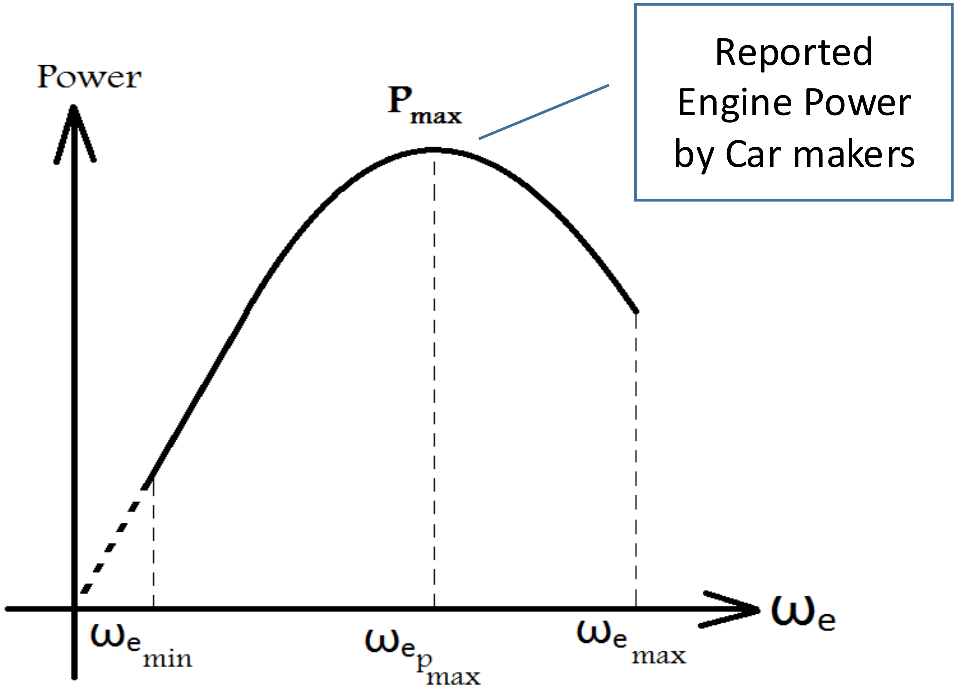
Typical torque curve for petrol engines



$$T_{e_{max}}(\omega_e) = A_0 + A_1\omega_e + A_2\omega_e^2$$

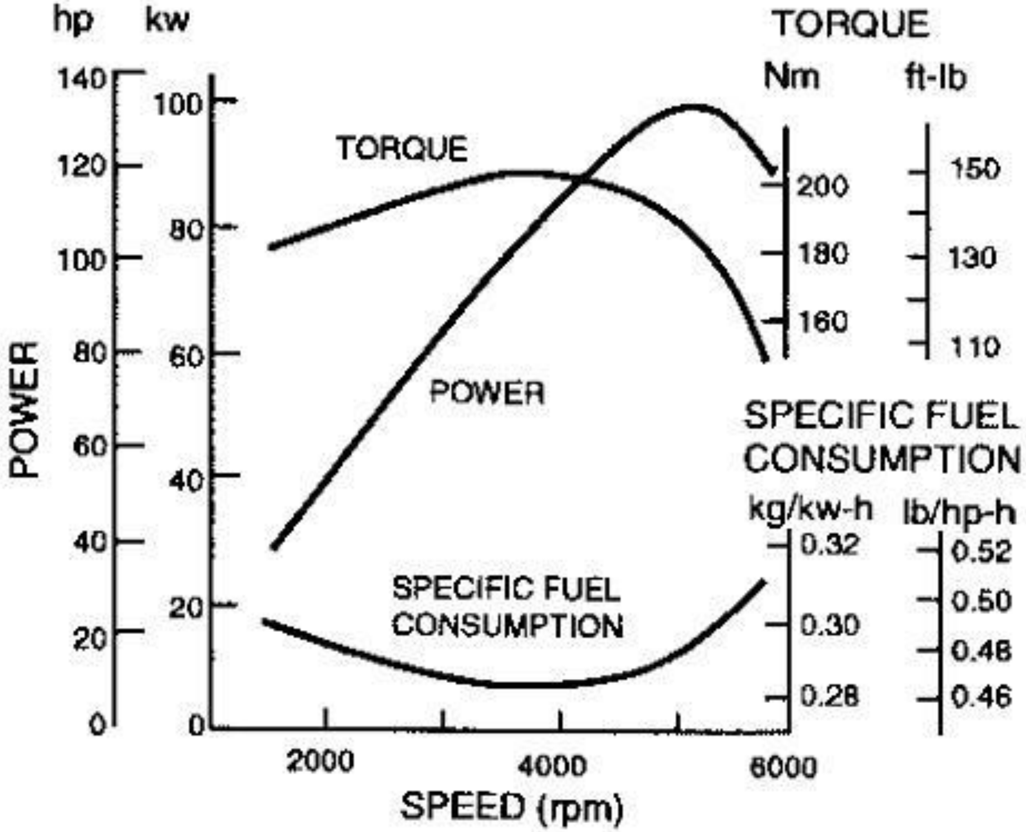
$$T_e = T_e(\omega_e, x_\theta) \approx x_\theta(A_0 + A_1\omega_e + A_2\omega_e^2)$$

Typical Power curve for petrol engines

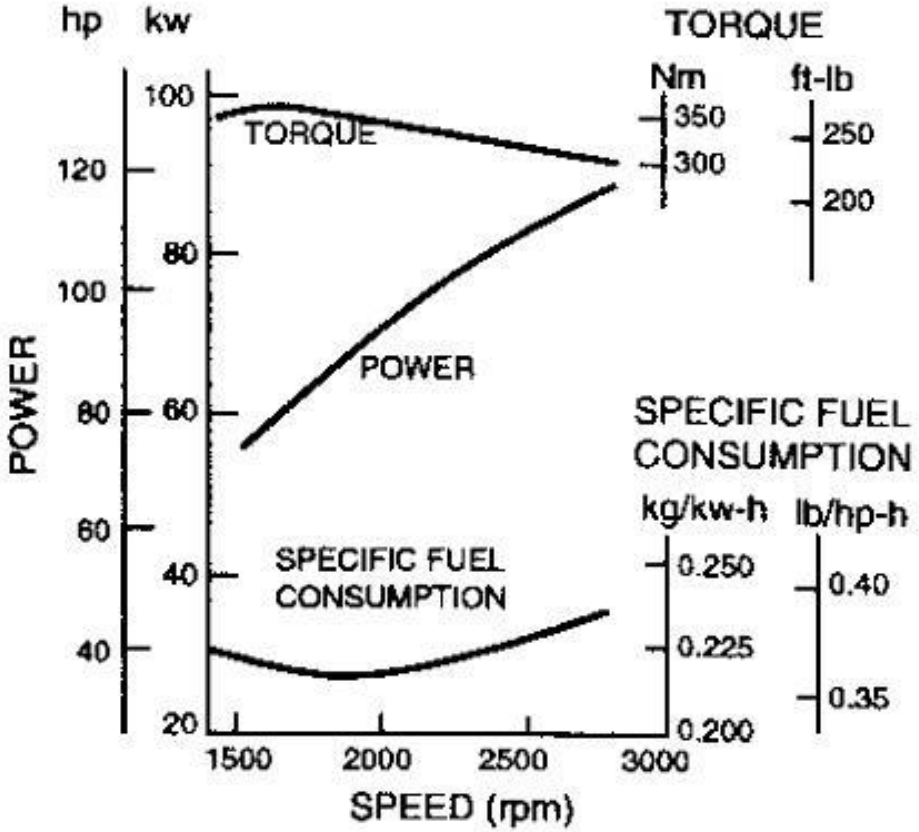


$$P_e(\omega_e, x_\theta) = \omega_e T_e(\omega_e, x_\theta) = x_\theta\omega_e(A_0 + A_1\omega_e + A_2\omega_e^2)$$

Accelerating Model: Engine Performance Characteristics

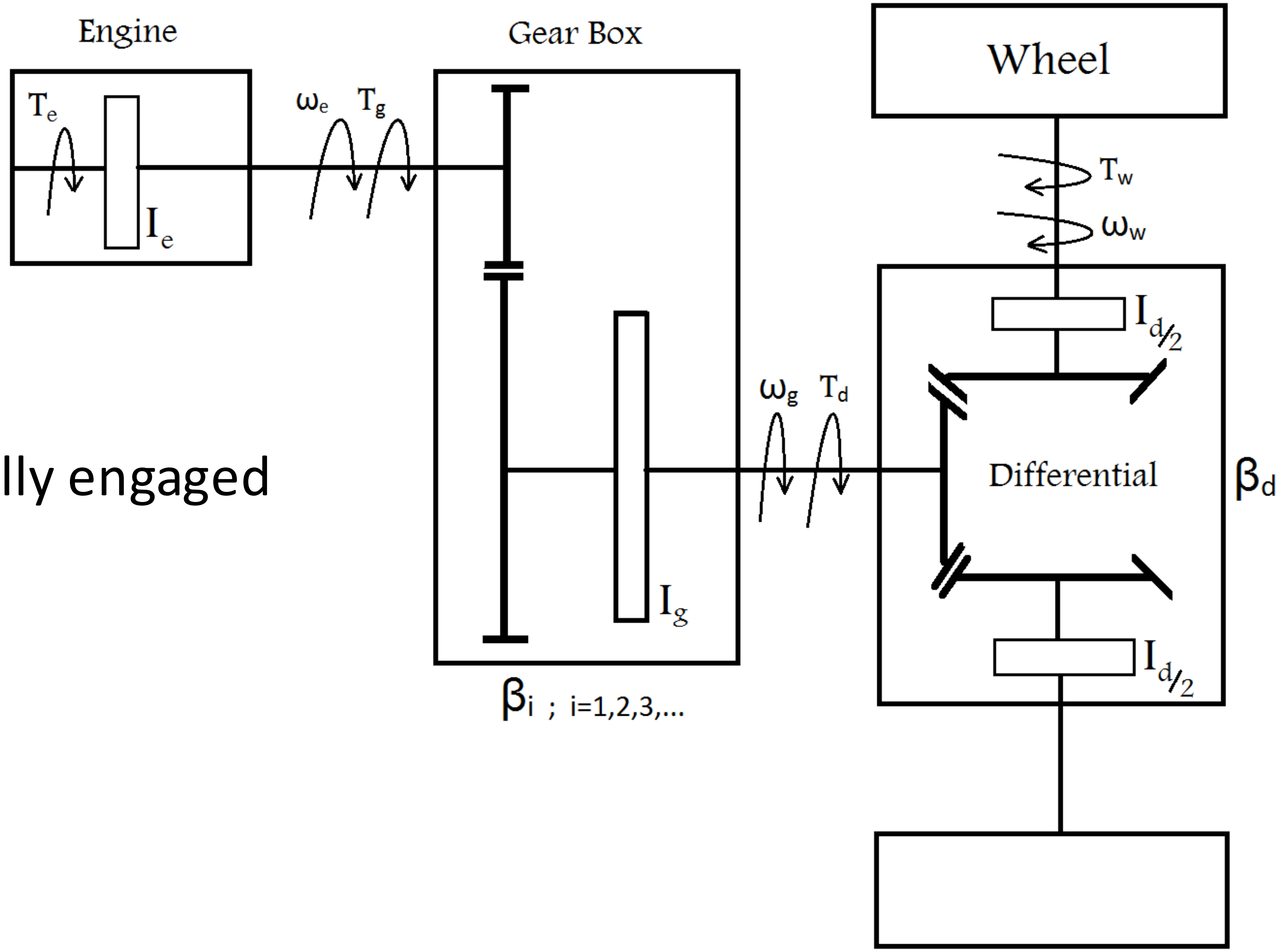


Gasoline



Diesel

Accelerating Model: Powertrain Model



Assumptions:

-2WD configuration

-Clutch/Torque converter always fully engaged

-Open differential

$$T_{w_l} = T_{w_R}$$

$$\omega_{w_l} \neq \omega_{w_R}$$

Accelerating Model: Powertrain Model

Engine's equation
$$T_e(\omega_e, x_\theta) - T_g = I_e \frac{d\omega_e}{dt} \qquad (1)$$

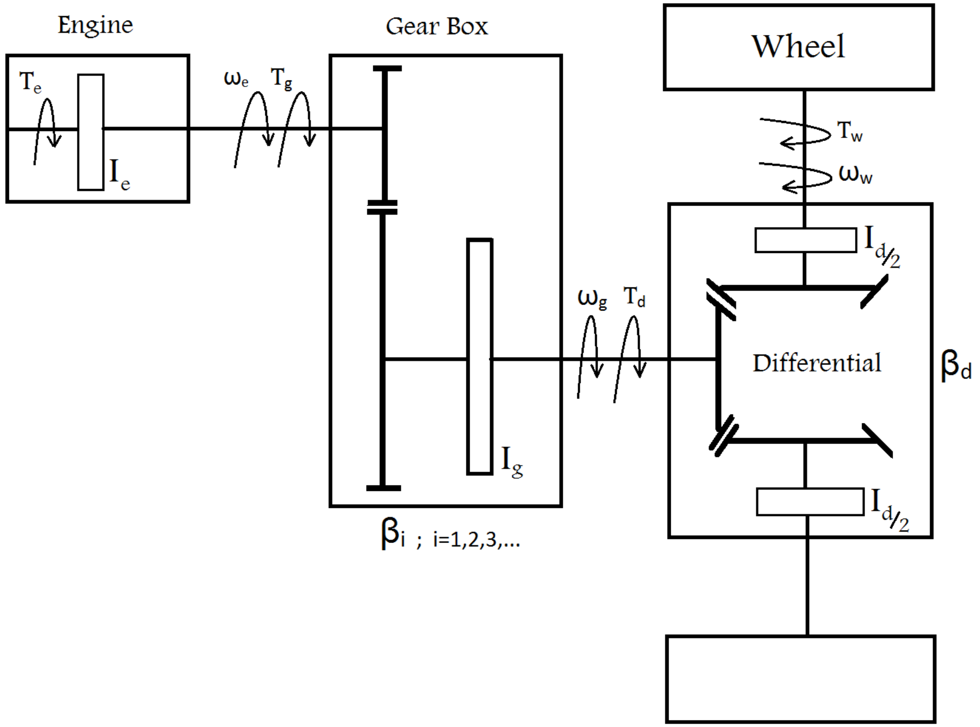
Gearbox's equation
$$\eta_g \beta_i T_g - T_d = I_g \frac{d\omega_g}{dt} \qquad \begin{cases} \omega_g = \dfrac{\omega_e}{\beta_i} \\ T_d = \beta_i T_g \end{cases} \qquad (2)$$

Differential's equations
$$\begin{cases} 0.5\eta_d \beta_d T_d - T_{wL} = 0.5 I_d \dfrac{d\omega_{wL}}{dt} \\ 0.5\eta_d \beta_d T_d - T_{wR} = 0.5 I_d \dfrac{d\omega_{wR}}{dt} \end{cases}$$

Open Differential $\qquad T_{wR} = T_{wL} = T_w$

Straight Line Driving $\qquad \omega_{wR} = \omega_{wL} = \omega$

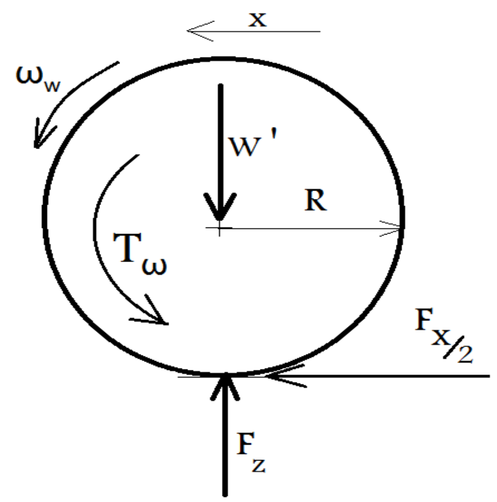
$$\eta_d \beta_d T_d - 2T_w = I_d \frac{d\omega_w}{dt} \qquad \omega_w = \frac{\omega_g}{\beta_d} \qquad (3)$$



Accelerating Model: Powertrain Model

Wheel's rotational dynamics equation

$$T_w - 0.5RF_x = I_w \frac{d\omega_w}{dt} \qquad (4)$$

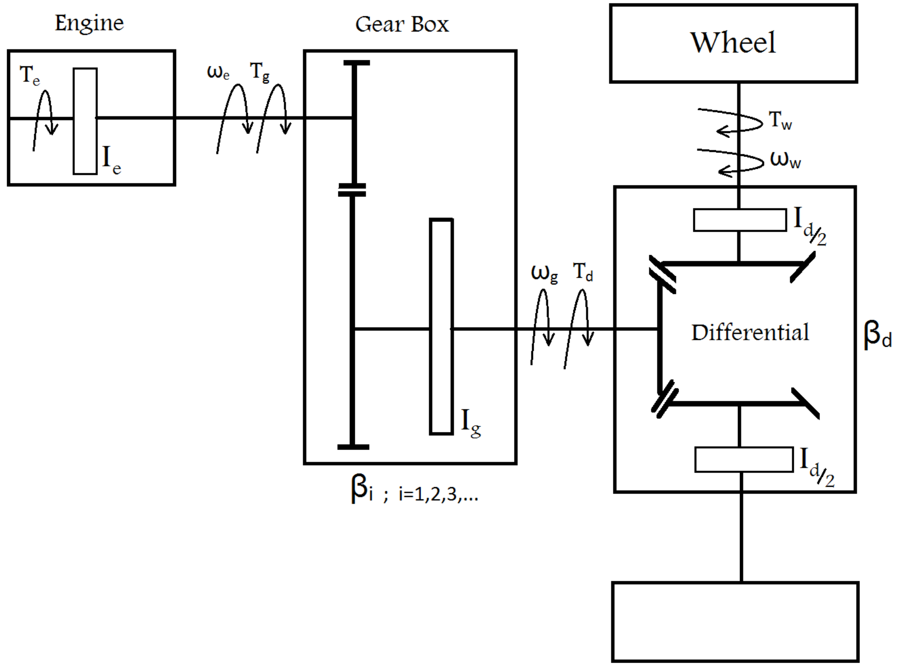


General form of vehicle longitudinal dynamics equation

$$F_x - F_R = m\frac{du}{dt} \qquad (5)$$

Assume longitudinal slip is very small:

$$u = R\omega_w \qquad (6)$$



Accelerating Model: Powertrain Model

(1) $\longrightarrow$ $$T_g = T_e - I_e \frac{d\omega_e}{dt}$$

(2) $\boxed{\eta_g \beta_i T_g - T_d = I_g \frac{d\omega_g}{dt}}$ $\boxed{\omega_g = \frac{\omega_e}{\beta_i}}$

$\longrightarrow$ $$T_d = \eta_g \beta_i T_e - \eta_g \beta_i^2 I_e \frac{d\omega_g}{dt} - I_g \frac{d\omega_g}{dt}$$

(3) $\boxed{\eta_d \beta_d T_d - 2T_w = I_d \frac{d\omega_w}{dt}}$ $\boxed{\omega_w = \frac{\omega_g}{\beta_d}}$

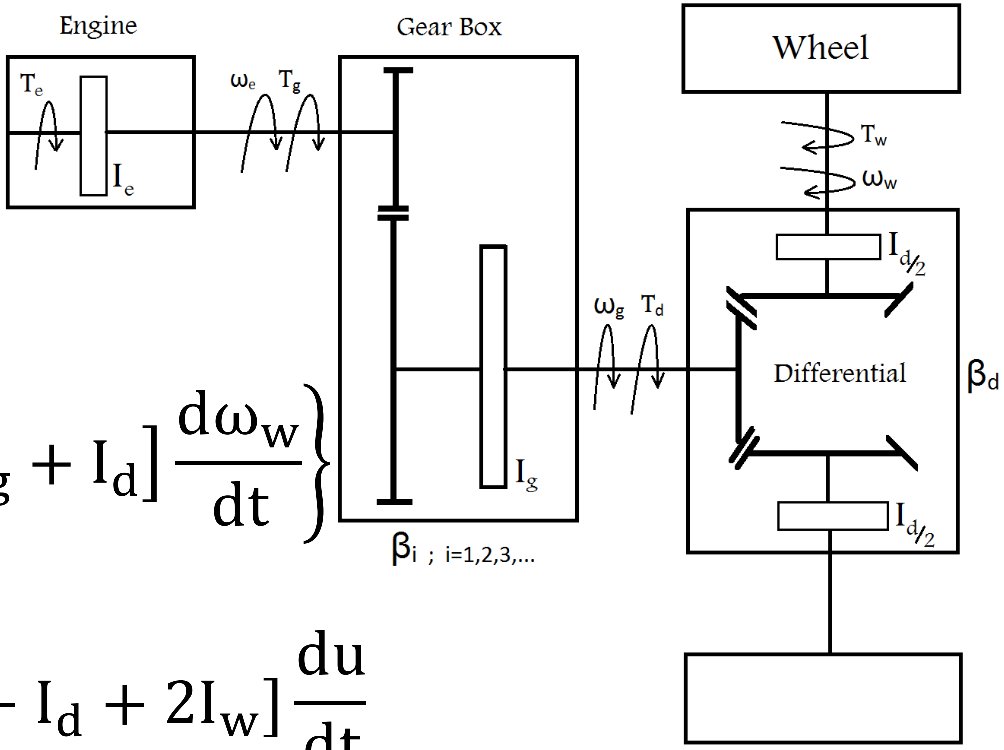
$\longrightarrow$ $$T_w = \frac{1}{2}\left\{\eta_g \eta_d \beta_i \beta_d T_e - [\eta_g \eta_d \beta_i^2 \beta_d^2 I_e + \eta_d \beta_d^2 I_g + I_d] \frac{d\omega_w}{dt}\right\}$$

(4&6) $\boxed{T_w - 0.5RF_x = I_w \frac{d\omega_w}{dt}}$ $\boxed{u = R\omega_w}$

$\longrightarrow$ $$F_x = \frac{\eta_g \eta_d \beta_i \beta_d}{R} T_e - \frac{1}{R^2}[\eta_g \eta_d \beta_i^2 \beta_d^2 I_e + \eta_d \beta_d^2 I_g + I_d + 2I_w] \frac{du}{dt}$$

(5) $\boxed{F_x - F_R = m \frac{du}{dt}}$

$\longrightarrow$ $$\frac{\eta_g \eta_d \beta_i \beta_d}{R} T_e - F_R = \underbrace{\left[m + \frac{1}{R^2}(\eta_g \eta_d \beta_i^2 \beta_d^2 I_e + \eta_d \beta_d^2 I_g + I_d + 2I_w)\right]}_{m_d:\ \text{Dynamic Vehicle Mass}} \frac{du}{dt}$$

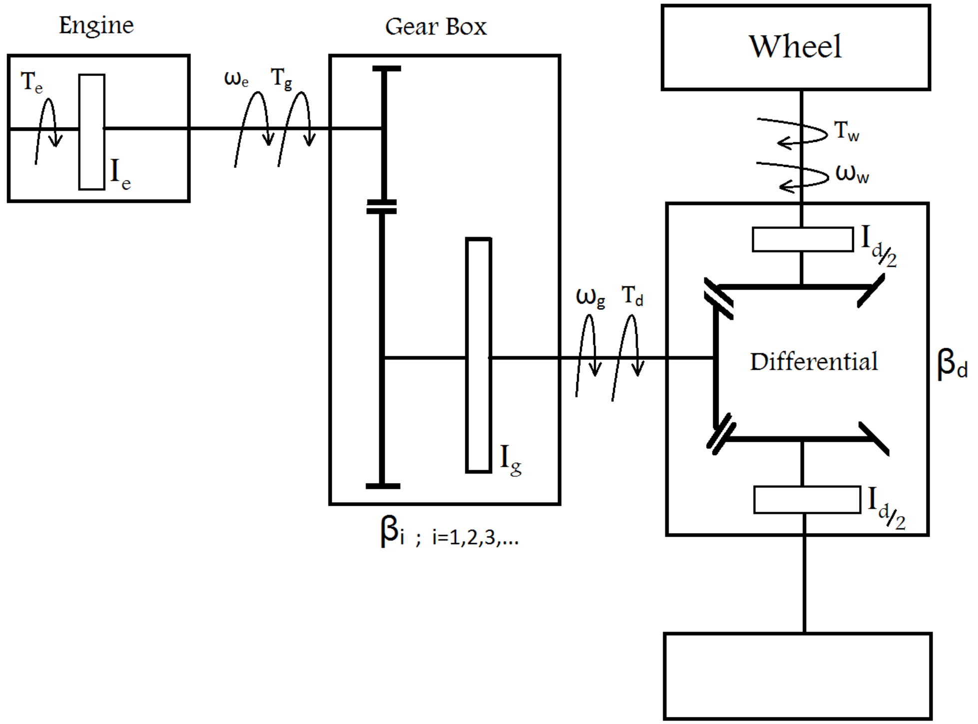


Accelerating Model: Powertrain Model

$$m_d = m + \frac{1}{R^2}\left(\eta_g\eta_d\beta_i^2\beta_d^2 I_e + \underbrace{\eta_d\beta_d^2 I_g + I_d + 2I_w}_{\text{Negligible Part}}\right) \approx m + \frac{1}{R^2}\eta_g\eta_d\beta_i^2\beta_d^2 I_e$$

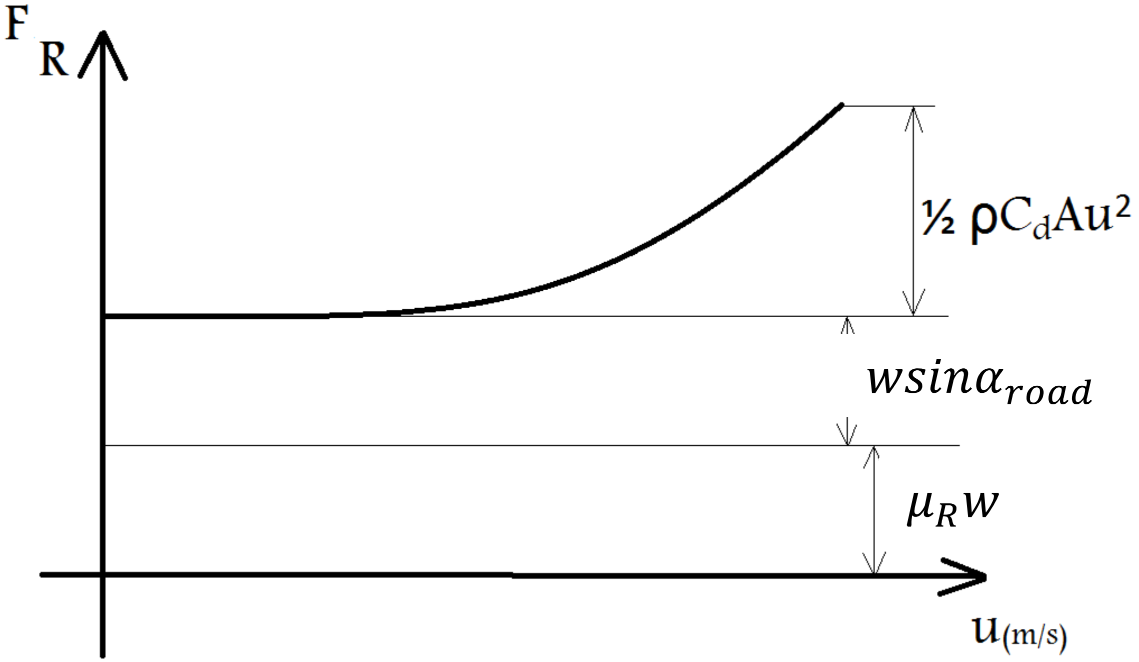
Final form of vehicle longitudinal dynamics equations:

$$\begin{cases} \underbrace{\frac{\eta_g\eta_d\beta_i\beta_d}{R} T_e(\omega_e, x_\theta)}_{\textit{Tractive Force}} - \underbrace{F_R}_{\textit{Resistance Force}} = m_d \frac{du}{dt} \\ m_d = m + \frac{\eta_g\eta_d\beta_i^2\beta_d^2}{R^2} I_e \\ \frac{\omega_e}{\beta_i\beta_d} R = u \end{cases}$$



Accelerating Model: Resistance Forces

$$F_R \begin{cases} \textbf{Rolling Resistance} \longrightarrow F_{RR} = \mu_R mg \\ \textbf{Gradient Resistance} \longrightarrow F_{RG} = mgsin\alpha_{road} \\ \textbf{Aerodynamics Resistance} \longrightarrow F_{R_A} = \frac{1}{2}\rho c_d A u^2 \end{cases}$$



$$F_R = \underbrace{wsin\alpha_{road} + \mu_R w + \frac{1}{2}\rho c_d A u^2}_{\mathrm{B_0} + \mathrm{B_1 u^2}}$$

$$w = mg,$$

Accelerating Model: Closed Form Formulation

$$F_t - F_R = m_d \frac{du}{dt}$$

$$F_t = \frac{\eta_t \beta_t T_e(\omega_e x_\theta)}{R}$$

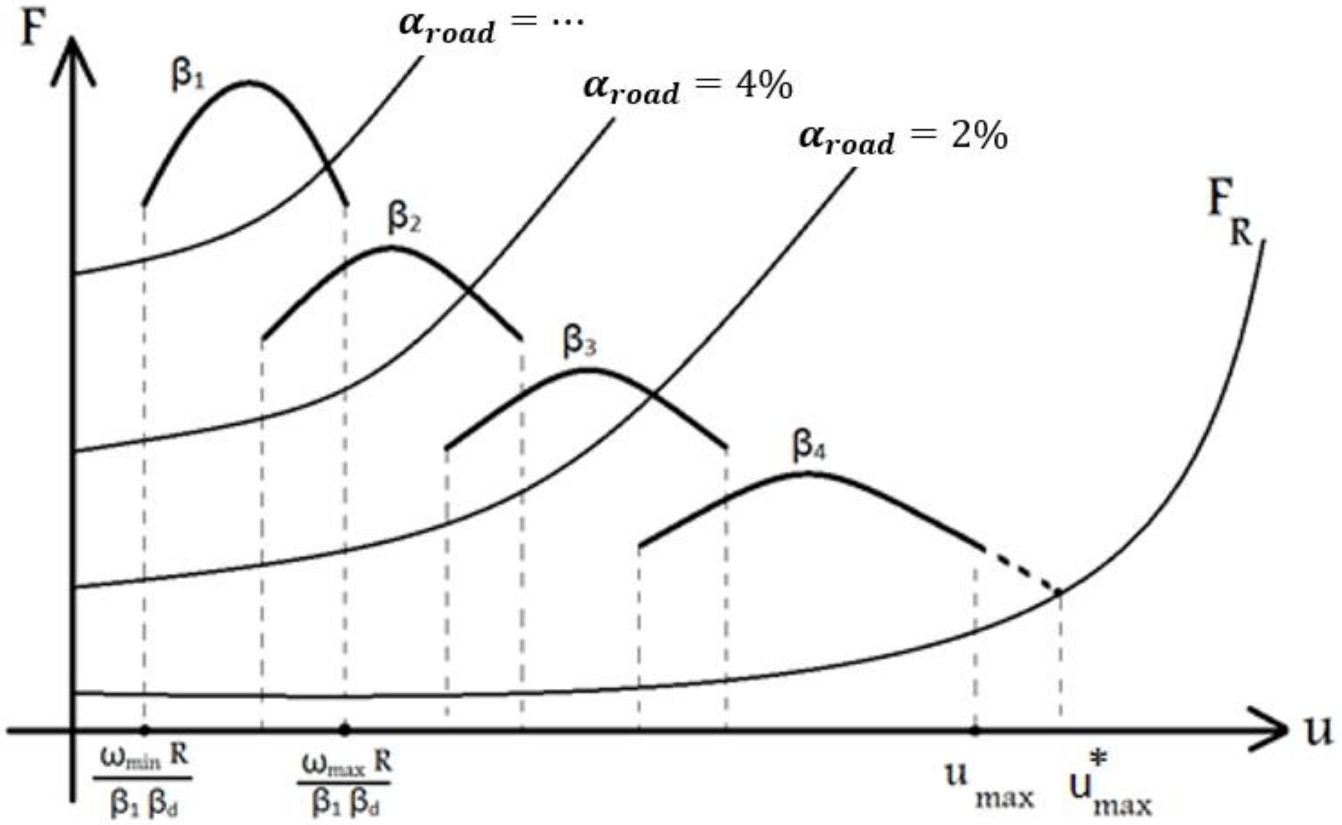
$$F_R = w(sin\alpha_{road} + \mu_R) + \frac{1}{2}\rho c_d A u^2$$

$$T_e = x_\theta(A_0 + A_1\omega_e + A_2\omega_e^2)$$

$$\omega_e = \frac{\beta_t}{R} u \qquad \beta_t = \beta_i \beta_d$$

$$C_0 + C_1 u + C_2 u^2 = m_d \frac{du}{dt}$$

Force -Velocity (F-u) Diagram



Example: Prediction of Maximum Velocity

$$C_0 + C_1 u + C_2 u^2 = m_d \cancelto{0}{\frac{du}{dt}}$$

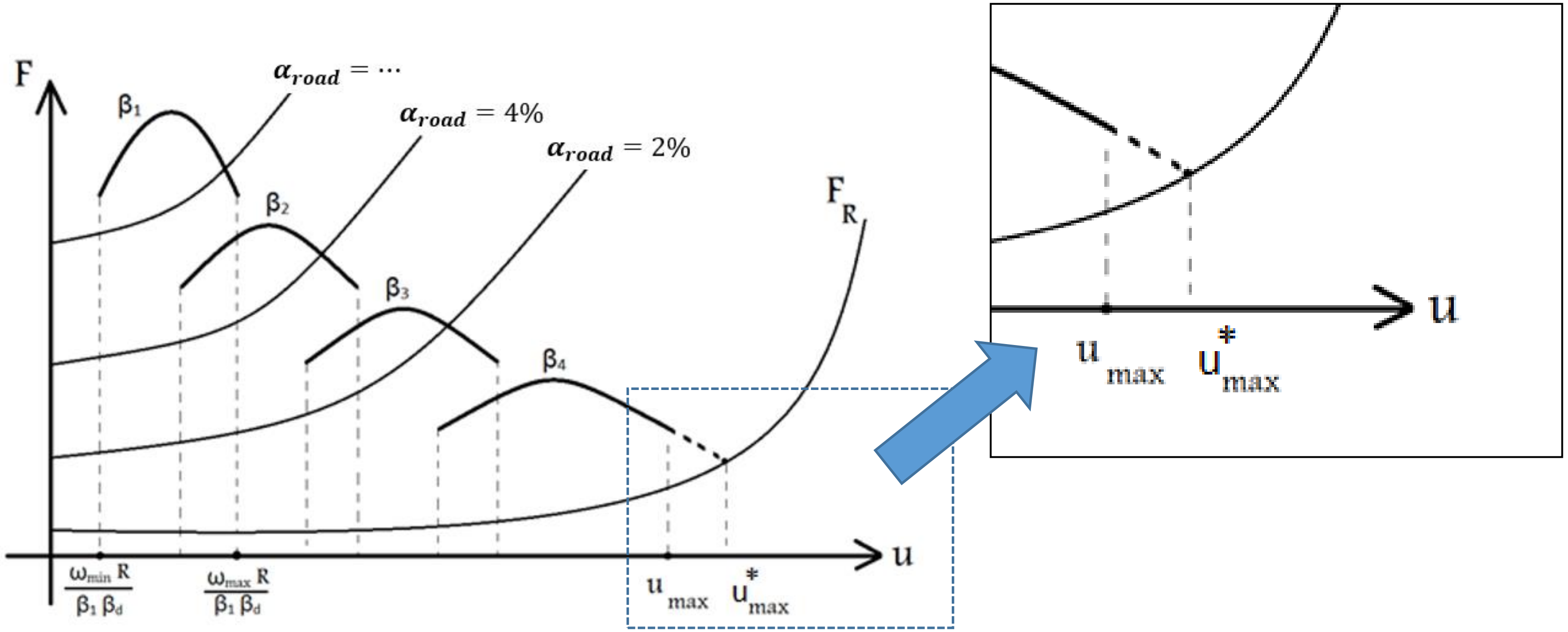
$$C_0 + C_1 u^*_{max} + C_2 u^{*2}_{max} = 0$$

Assumptions:

- Full Throttle $\longrightarrow$ $x_\theta = 100\%$
- Flat road $\longrightarrow$ $\alpha_{road} = 0$
- Minimum Vehicle Weight $\longrightarrow$ $m = m_{min}$
- Lowest Gear Ratio $\longrightarrow$ $\beta_i = \beta_{min}\ , i = max$

Example: Prediction of Maximum Velocity

Maximum Engine RPM Constrain:



$$
\begin{cases}
\textit{if: } \omega_{e_{u_{max}}} \le \omega_{e_{max}} \quad \textit{then} \quad u_{max} = u^*_{max} \\
\qquad \qquad \qquad \frac{u_{max}}{R}\beta_{min}\beta_d \\
\textit{else: } \quad u_{max} = \frac{\omega_{e_{max}}}{\beta_{min}\beta_d} R
\end{cases}
$$

Example: Maximum Gradeability

Minimum expected gradeability
for passenger cars

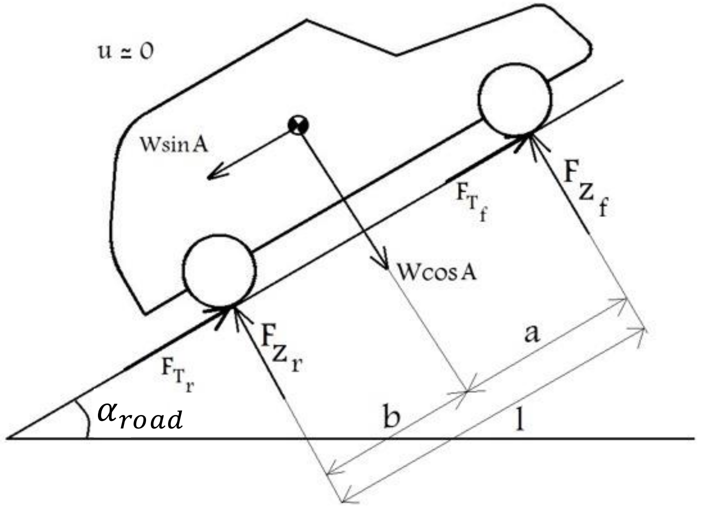
$\tan\alpha_{road} = 33\% \longrightarrow \alpha_{road} \approx 18^{\circ}$

Maximum gradeability of off-roaders
is considerably higher

$\tan\alpha_{road} = 100\% \longrightarrow \alpha_{road} \approx 45^{\circ}$

Assumptions:

- Maximum Engine Torque $\longrightarrow$ $\mathrm{T}_e = \mathrm{T}_{e_{\max}}$
- Very low constant velocity $\longrightarrow$ $\dot{u}, u \approx 0$
- Maximum Vehicle Weight $\longrightarrow$ $m = m_{max}$
- Highest Gear Ratio $\longrightarrow$ $\beta_1 = \beta_{\max}$, $i = 1$



Example: Maximum Gradeability

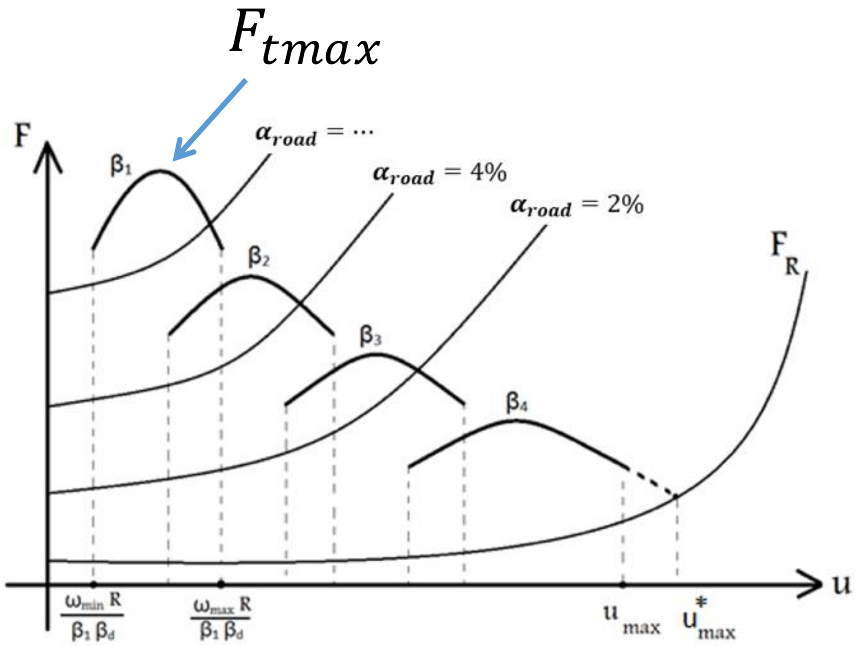
$$F_{tmax} - F_R = 0$$

$$\beta_1 \beta_d \qquad F_{RG} + F_{RR} + \cancel{F_{RA}} \qquad 0$$

$$F_{t_{max}} = \frac{\eta_t \beta_{t_{max}} T_{e_{max}}}{R}$$

$$\frac{\eta_t \beta_{t_{max}} T_{e_{max}}}{R} = w_{max} sin\alpha^*_{road_{max}} + w_{max}\mu_R$$

$$\alpha^*_{road_{max}} = sin^{-1}[\frac{\eta_t \beta_{t_{max}} T_{e_{max}}}{R w_{max}} - \mu_R]$$

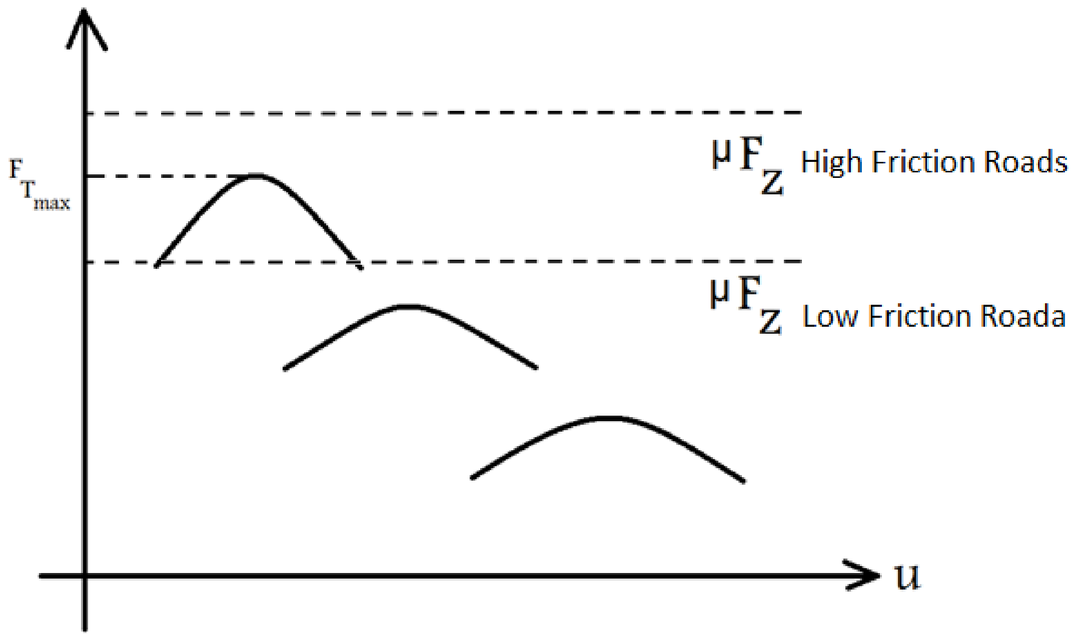


Example: Maximum Gradeability

Road Friction Capacity Constrain

$$F_{t_{max}} \leq \mu_{Road}\, F_{z\, Drive\ Axle(s)}$$

- The calculated gradeability is valid if the maximum generated tractive force is less than the road frictional capacity
- In some cases, such as driving on a low friction road, the vehicle's tractive force can be higher than the road capacity.
- In such cases, the road capacity determines the maximum gradeability rather than the powertrain torque.

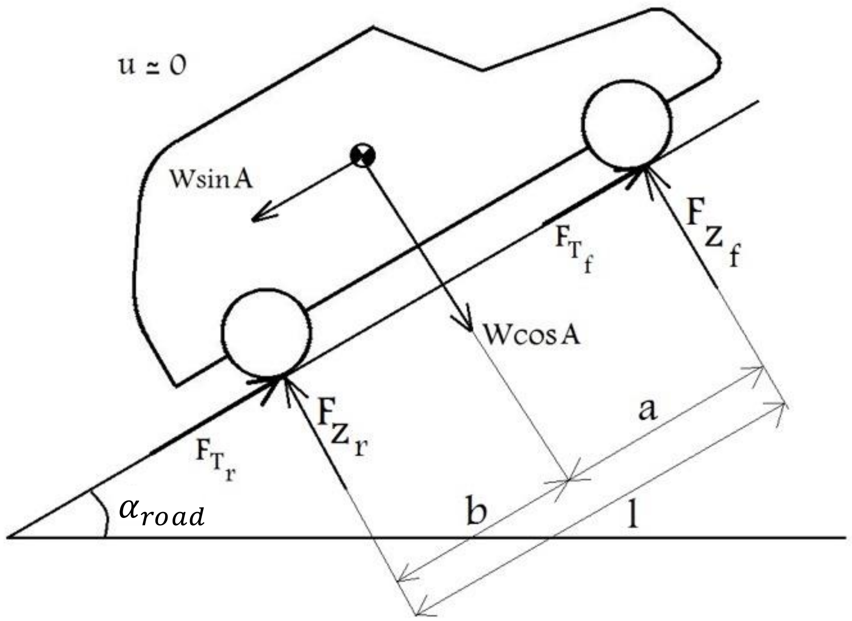


Example: Maximum Gradeability

Calculation of the Road Friction Capacity

Calculation of Normal Forces:

$$\begin{cases} f_{z_f} + f_{z_r} = w\cos\ \alpha_{road} \\ f_{z_f}.l + \ wsin\alpha_{road}.h - wcos\alpha_{road}.b = 0 \end{cases}$$



$$\Rightarrow \begin{cases} f_{z_f} = w\left(\boxed{-\frac{h}{l}sin\alpha_{road}} + \frac{b}{l}cos\alpha_{road}\right) \\ f_{z_r} = w\Big(\underbrace{\boxed{\frac{h}{l}sin\alpha_{road}}}_{\text{Load Transfer}} + \frac{a}{l}cos\alpha_{road}\Big) \end{cases}$$

Example: Maximum Gradeability

Calculation of the Road Friction Capacity

$$FWD\ :\ F_{x_f} = \mu_{road} f_{z_f} = w\mu_{road}(-\frac{h}{l}\sin\alpha_{road} + \frac{b}{l}\cos\alpha_{road})$$

$$RWD\ :\ F_{x_f} = \mu_{road} f_{z_r} = w\mu_{road}(\frac{h}{l}\sin\alpha_{road} + \frac{b}{l}\cos\alpha_{road})$$

$$4WD\ :\ F_{x_f} = \mu_{road}(f_{z_r} + f_{z_f}) = w\mu_{road}\cos\alpha_{road}$$

$$\boxed{F_{x_{f4wd}} > F_{x_{fRWD}} > F_{x_{fFWD}}}$$

Example: Maximum Gradeability

Calculation of Maximum Gradeability based on the Road Friction Capacity

For a FWD vehicle: $F_{x_{fFwd}} - F_R = 0$

$$\underbrace{w_{max}\mu_{road}\left(-\frac{h}{l}\sin\alpha'_{road_{max}} + \frac{b}{l}\cos\alpha'_{road_{max}}\right)}_{F_{x_{fwd}}} - \underbrace{w_{max}\mu_R}_{F_{R_R}} - \underbrace{w_{max}\sin\alpha'_{road_{max}}}_{F_{R_G}} = 0$$

$$-(1+\mu_{road}\frac{h}{l})\sin\alpha'_{road_{max}} + \mu_{road}\frac{b}{l}\cos\alpha'_{road_{max}} = \mu_R$$

Rolling resistance is much smaller than gradient resistance, so it can be ignored:

$$-(1+\mu_{road}\frac{h}{l})\sin\alpha'_{road_{max}} + \mu_{road}\frac{b}{l}\cos\alpha'_{road_{max}} = 0$$

Example: Maximum Gradeability

Calculation of Maximum Gradeability based on the Road Friction Capacity

$$\begin{cases} FWD: \alpha'_{\substack{road\,max \\ FWD}} = tan^{-1}\left[\dfrac{\mu_{road}\dfrac{b}{l}}{1+\mu_{road}\dfrac{h}{l}}\right] \\ RWD: \alpha'_{\substack{road\,max \\ RWD}} = tan^{-1}\left[\dfrac{\mu_{road}\dfrac{a}{l}}{1-\mu_{road}\dfrac{h}{l}}\right] \\ 4WD: \alpha'_{\substack{road\,max \\ 4WD}} = tan^{-1}[\mu_{road}] \end{cases}$$

$$\boxed{-(1+\mu_{road}\frac{h}{l})sin\,\alpha'_{road_{max}} + \mu_{road}\frac{b}{l}cos\,\alpha'_{road_{max}} = 0}$$

Final Value of Maximum Gradeability

$$\alpha_{road\,max} = Min\left(\alpha^*_{road_{max}}, \alpha'_{road_{max}}\right)$$

Example: Acceleration Time

Different Standard Tests for Evaluation of vehicle's Acceleration Performance:

- t: 0 $\longrightarrow$ 100km/h (sec)
- t: 60 $\longrightarrow$ 100km/h (sec) $\longrightarrow$ End Acceleration
- t: 0 $\longrightarrow$ 50km/h (sec) $\longrightarrow$ Starting Acceleration
- t: 0 $\longrightarrow$ 400m
- V: 0 $\longrightarrow$ 5m/s

Example: Acceleration Time

Prediction of Acceleration Time

Assumptions:

- Full Throttle $\longrightarrow$ $x_\theta = 100\%$
- Flat road $\longrightarrow$ $\alpha_{road} = 0$
- Minimum Vehicle Weight $\longrightarrow$ $m = m_{min}$
- Expert Driving

General form of the longitudinal dynamics equation:

$$C_0 + C_1 u + C_2 u^2 = m_d \frac{du}{dt}$$

(see slide 16)

Where $\mathrm{C_0, C_1, C_2}$ and $\mathrm{m_d}$ are functions of gearbox gear ratio β_i

Example: Acceleration Time

Integration of longitudinal dynamics equation of motion:

$$dt = \frac{m_d du}{C_0 + C_1 u + C_2 u^2} \qquad \mathrm{u_1 \le u < u_2}$$

$$\mathrm{t} = \int_0^{\mathrm{t}} \mathrm{dt} = \int_{\mathrm{u_1}}^{\mathrm{u_2}} \frac{m_d du}{C_0 + C_1 u + C_2 u^2}$$

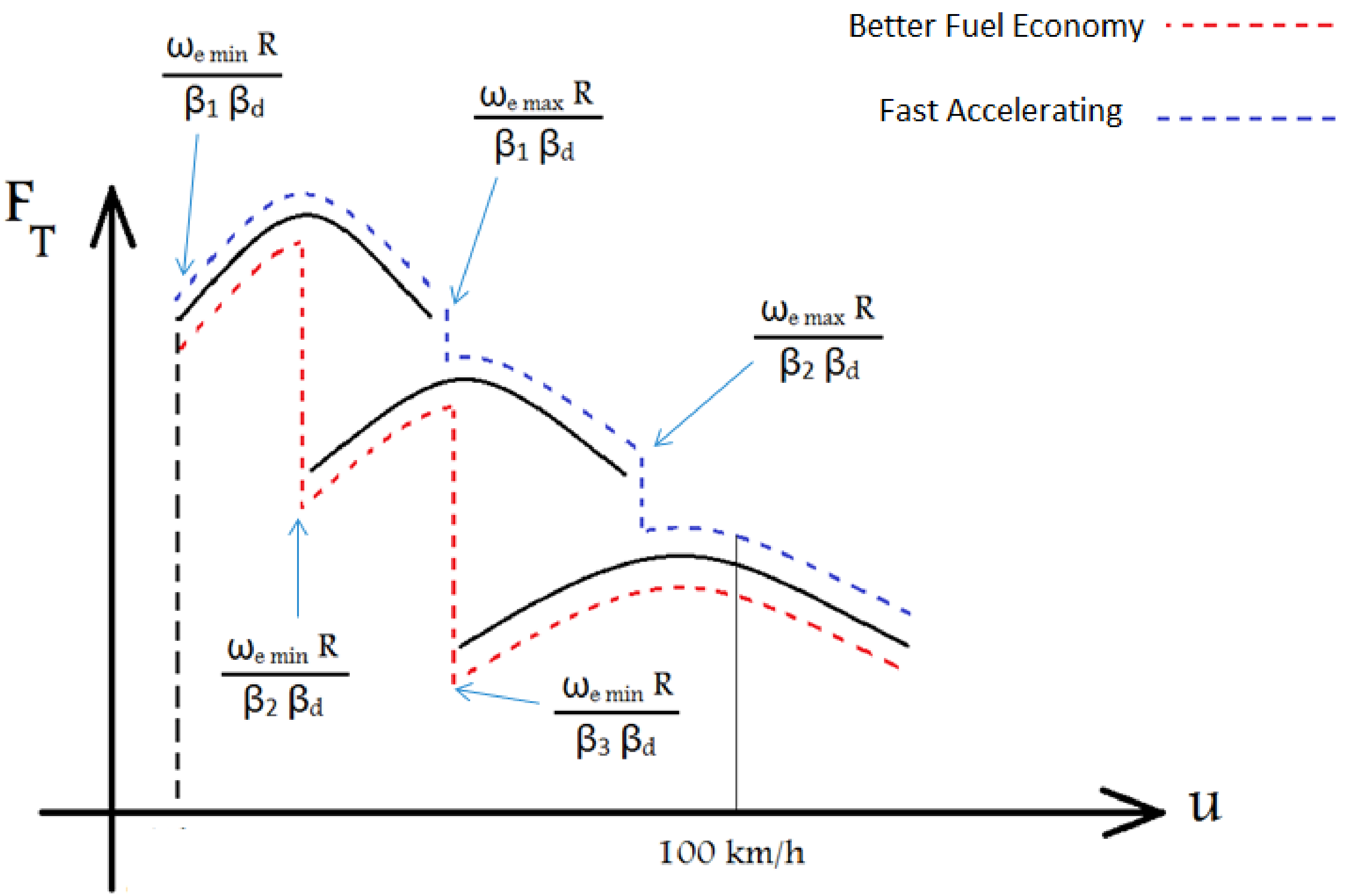
During the integration interval $u_1 \le u < u_2$, gearbox ratio should be constant. When the gear changes, the interval must be divided into constant gear ratio sub-intervals as follows:

$$\mathrm{t} = \mathrm{t_{Delay}} + \sum_i \int_{\mathrm{u_i}}^{\mathrm{u_{i+1}}} \frac{m_d du}{C_0 + C_1 u + C_2 u^2}$$

Gear Changing Time

Example: Acceleration Time

Effect of gear shifting strategy

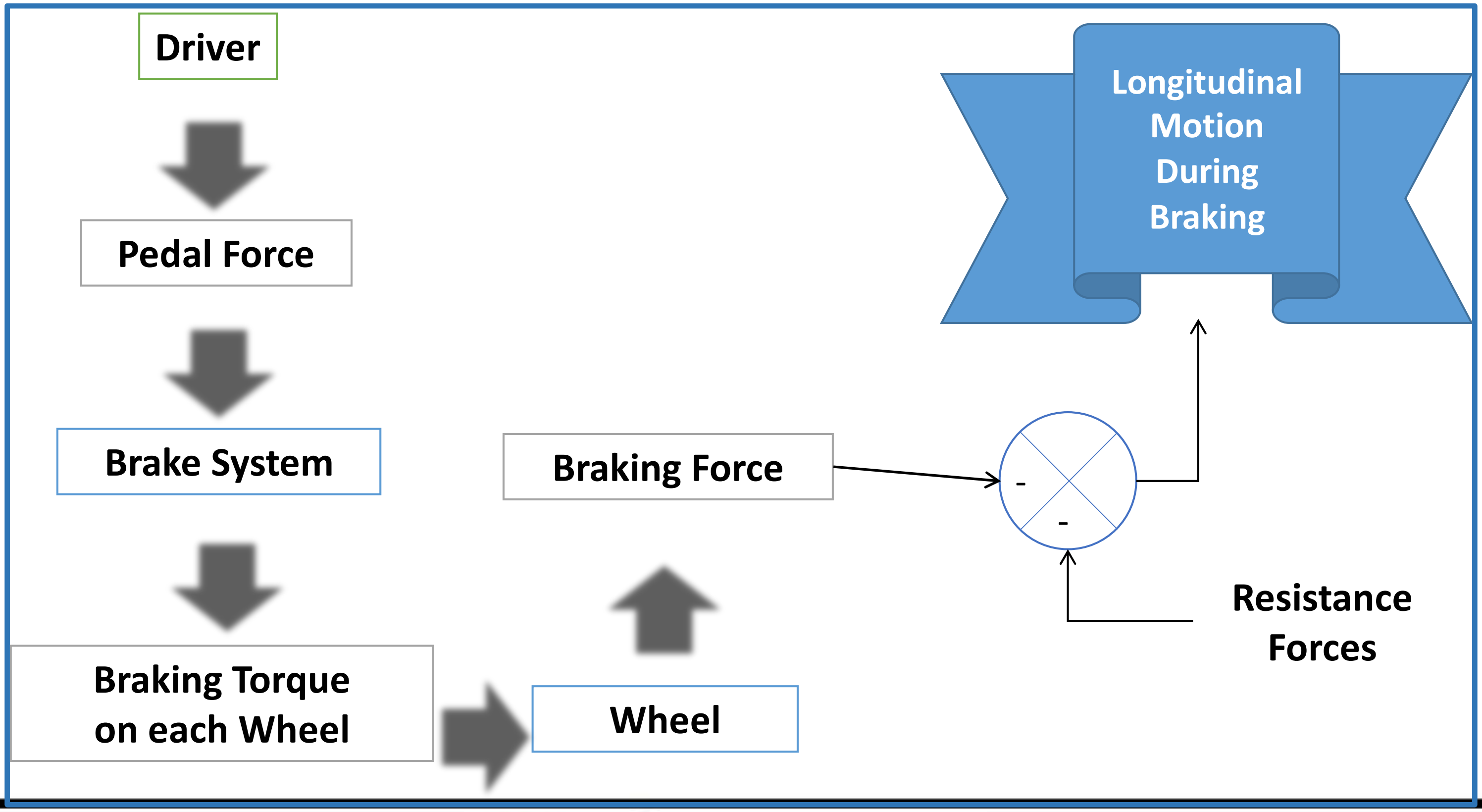


(see slide 16)

Braking Model

- **Modeling Objectives:**
  - **Evaluation of vehicle dynamics behavior during a straight-line braking.**
  - **Prediction of stopping distance in various road conditions.**

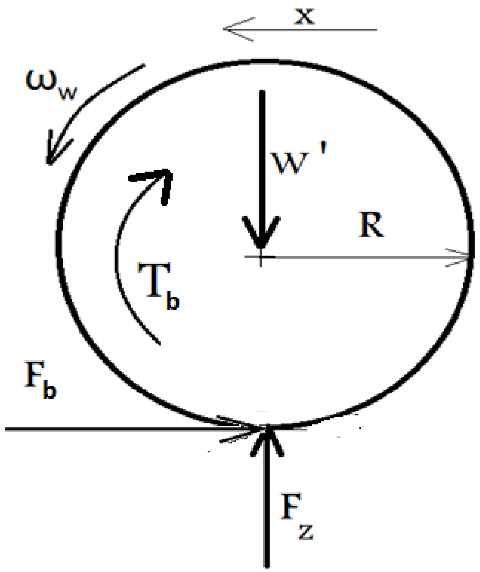
Braking Process



Braking Model

Wheel's rotational dynamics equation for wheel *i:*

$$-T_{b_i} - F_{b_i} R = I_{w_i} \frac{d\omega_{w_i}}{dt} \qquad i = fr, fl, rr, rl \tag{1}$$



Vehicle longitudinal dynamics equation during braking:

$$-\sum_i F_{bi} - F_R = m\frac{du}{dt} \tag{2}$$

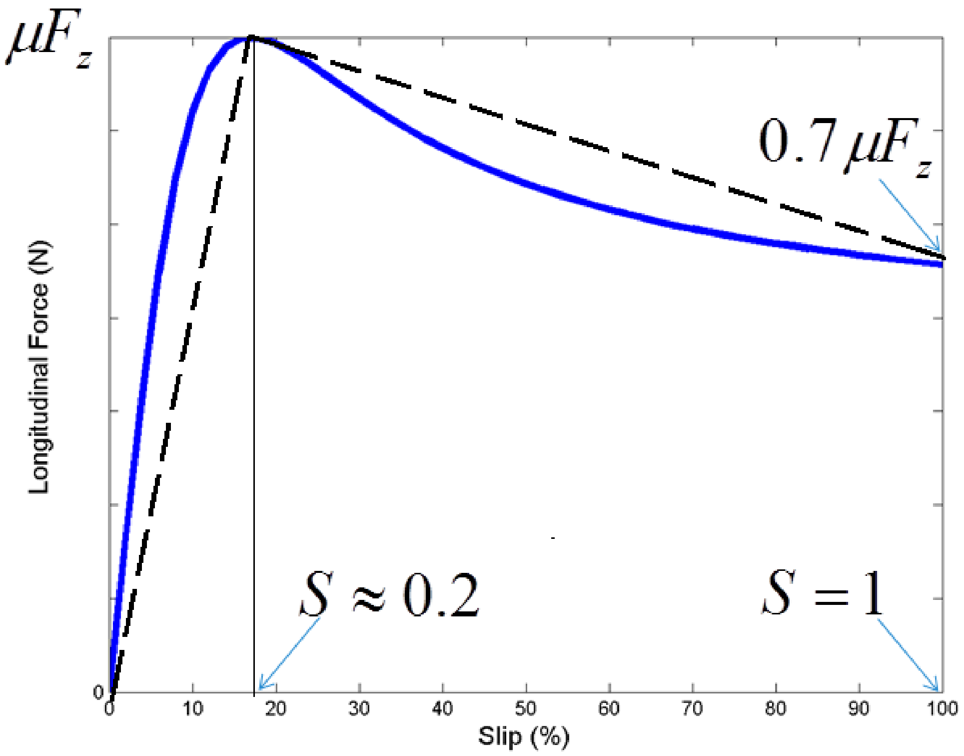
Based on definition of longitudinal slip:

$$S_i = 1 - \frac{R\omega_{w_i}}{u} \qquad i = fr, fl, rr, rl \tag{3}$$

Braking Model

Longitudinal Forces are calculated based on pure slip tire model :

$$F_{b_i} = F_{b_i}(S, F_{z_i}) \tag{4}$$



The approximated model (dashed line) can be used instead of exact tire model (solid line) with an acceptable accuracy:

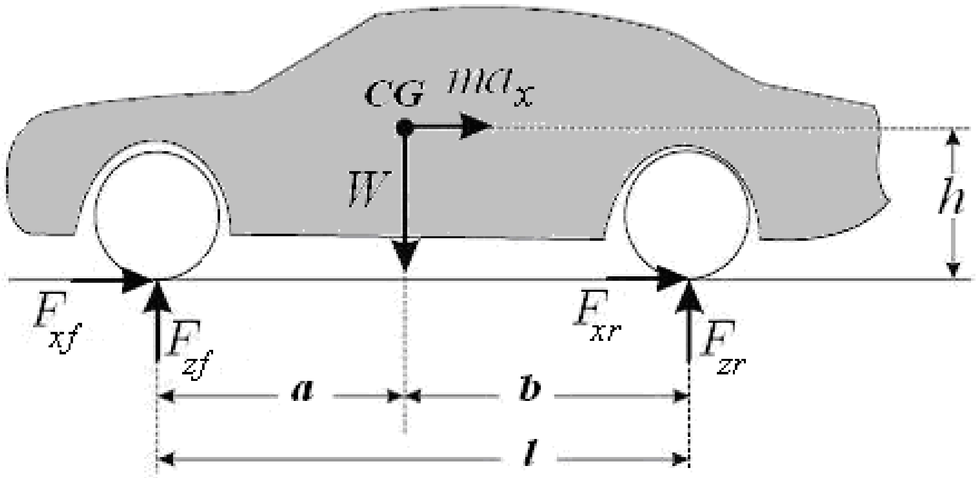
$$F_{b_i} = \begin{cases} 5S_i \mu F_{z_i} & S \le 0.2 \\ \dfrac{\mu F_{z_i}}{80}(-30S_i + 86) & S > 0.2 \end{cases} \tag{5}$$

Braking Model

Calculation of normal force of each wheel considering longitudinal load transfer due to braking:

$$F_{z_{rl}} = F_{z_{rr}} = \frac{W}{2l}\left(a + \frac{a_x}{g}h\right)$$

$$F_{z_{fl}} = F_{z_{fr}} = \frac{W}{2l}\left(b - \frac{a_x}{g}h\right) \qquad (6)$$



where: $a_x = \frac{du}{dt}$

The stopping distance will be calculated by using equations (2), (5), and (6) and:

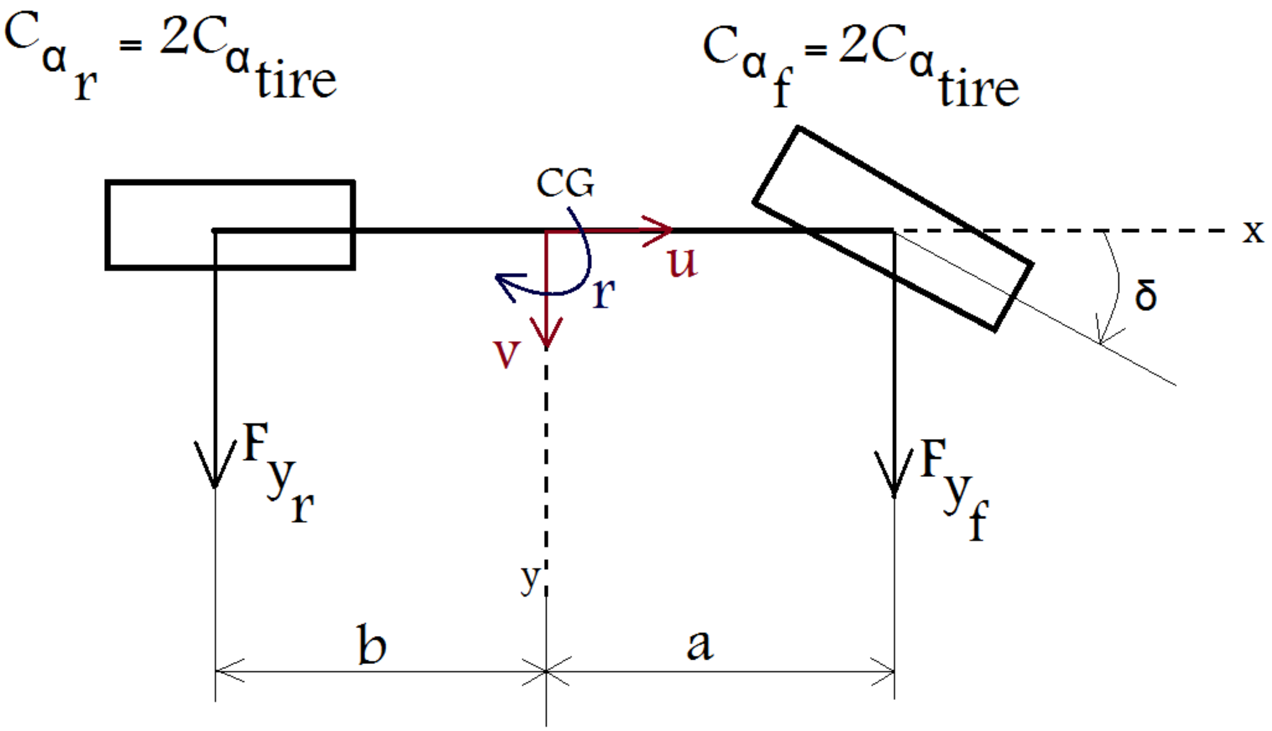
$$X = \int u\,dt \qquad (7)$$

Single-track Linear Handling Model

- **Modeling Objectives:**
  - Simulation of vehicle handling behavior during low-g maneuvers (regular driving) .
  - Having an analytical model for performing stability, steady state and transient analysis.
- **Modeling Assumptions:**
  - Planar motion, no roll and pitch motions
  - Driving on even road, no vertical motion
  - Linear tire model
  - Constant longitudinal velocity
  - Small sideslip angle
  - No load transfer
  - No camber change
  - Two degrees of freedom (lateral and yaw motions)

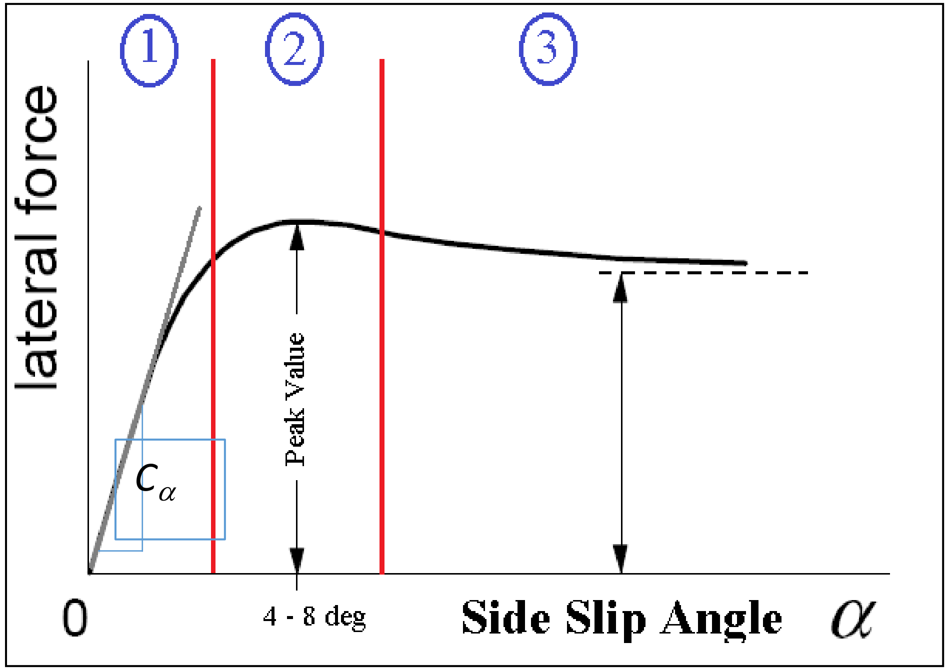
Single-track Model: Configuration

- **Model Configuration:**



Linear Tire Range

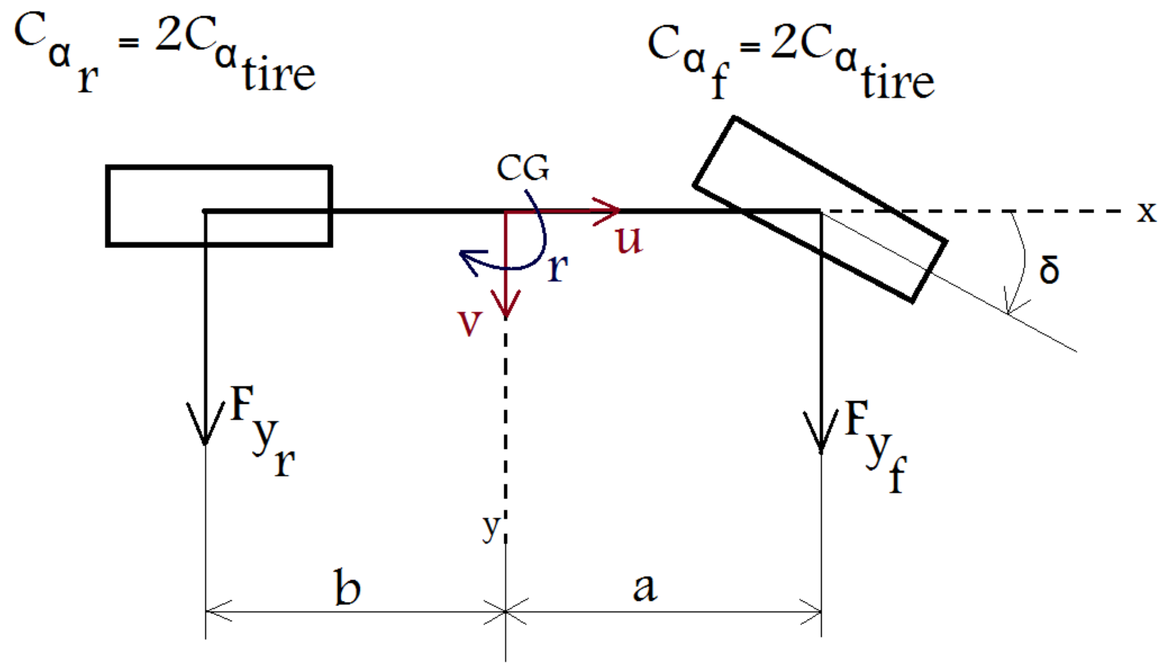
$F_y = C_\alpha \alpha \qquad \alpha < 4\ deg$



Single-track Model: Mathematical Representation

$$\begin{cases} \sum F_y = m(\dot{v} + ur) \\ \sum M_z = I_z \dot{r} \end{cases} \quad (1)$$

$$\begin{cases} F_{y_r} + F_{y_f} = m(\dot{v} + ur) \\ F_{y_f}.a - F_{y_r}.b = I_z \dot{r} \end{cases} \quad (2)$$



Single-track Model: Tire Forces

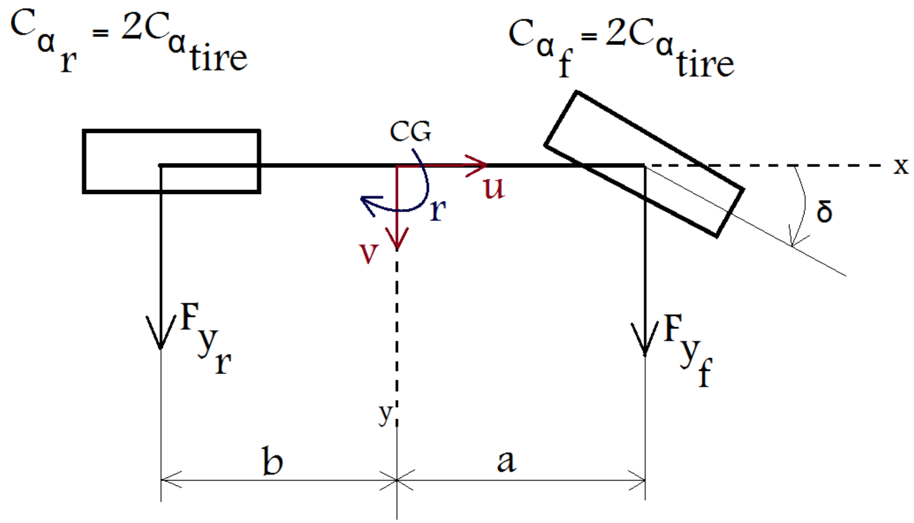
- **Mathematical Representation:**

Tire Model:

$$\begin{cases} F_{y_f} = c_{\alpha_f} . \alpha_f \\ F_{y_r} = c_{\alpha_r} . \alpha_r \end{cases} \qquad (3)$$

Front and rear tires sideslip angles:

$$\begin{cases} \alpha_f = \delta - \dfrac{v_f}{u_f} = \delta - \dfrac{v + ar}{u} \\ \alpha_r = 0 - \dfrac{v_r}{u_r} = -\dfrac{v - br}{u} \end{cases} \qquad (4)$$



Single-track Model: State-Space Model

- Combining equations 1-4, the closed-form representation of the model in state-space form can be found as:

$$F_{y_f} = c_{\alpha_f} . \alpha_f \qquad \alpha_f = \delta - \frac{v_f}{u_f} = \delta - \frac{v + ar}{u}$$

$$F_{y_r} = c_{\alpha_r} . \alpha_r \qquad \alpha_r = 0 - \frac{v_r}{u_r} = -\frac{v - br}{u}$$

$$\sum F_y = m(\dot{v} + ur)$$

$$\sum M_z = I_z \dot{r}$$

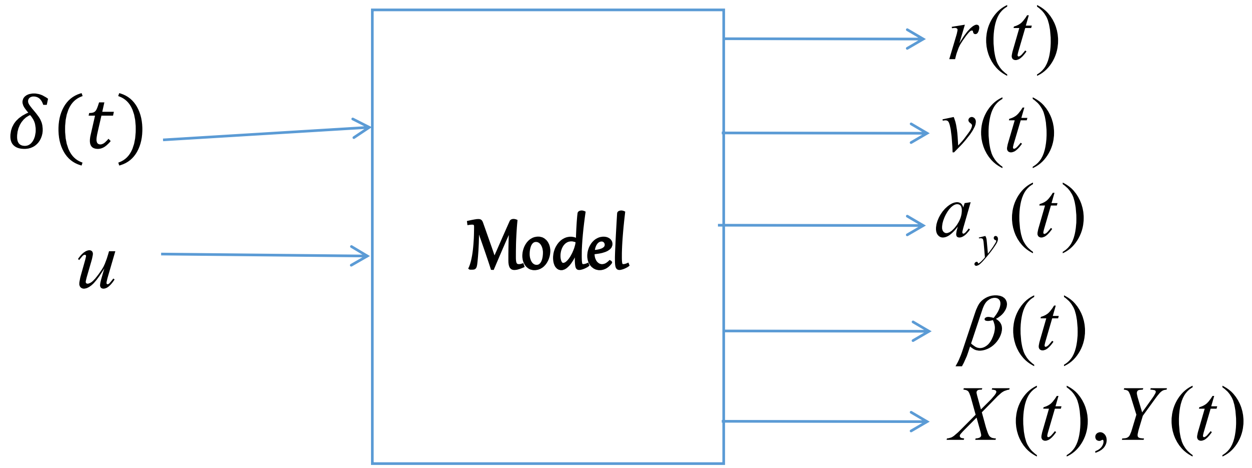
$$\dot{X} = AX + B\delta$$

$$\begin{bmatrix} \dot{v} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} -\dfrac{c_{\alpha_f} + c_{\alpha_r}}{um} & -(\dfrac{ac_{\alpha_f} - bc_{\alpha_r}}{um} + u) \\ -\dfrac{ac_{\alpha_f} - bc_{\alpha_r}}{uI_z} & -\dfrac{a^2 c_{\alpha_f} + b^2 c_{\alpha_r}}{uI_z} \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} + \begin{bmatrix} \dfrac{c_{\alpha_f}}{m} \\ \dfrac{ac_{\alpha_f}}{I_z} \end{bmatrix} \delta$$

$$a_y = \dot{v} + ur \qquad \beta = -\tan^{-1} \frac{v}{u}$$

Single-track Model for Vehicle Handling Analysis

- Simulation of vehicle handling using single-track Linear Handling Model (bicycle model):



$$a_y = \dot{v} + ur$$

$$\beta = -\tan^{-1}\frac{v}{u}$$

$$\mathrm{X(t)} = \int_0^t \dot{\mathrm{X}}\,(\mathrm{t})\mathrm{dt}$$

$$\psi = \int_0^t r(t)dt$$

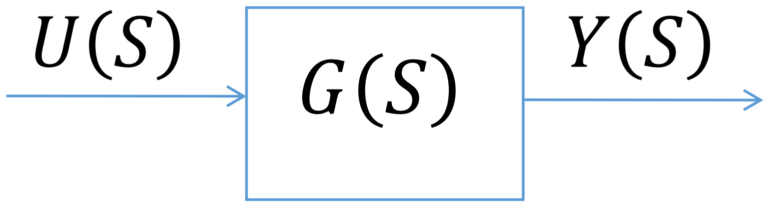
$$\dot{X}(t) = u\cos\psi - v\sin\psi$$

$$\dot{Y}(t) = u\sin\psi + v\cos\psi$$

$$\mathrm{Y(t)} = \int_0^t \dot{\mathrm{Y}}\,(\mathrm{t})\mathrm{dt}$$

Single-track Model: Stability Analysis

- **Background**

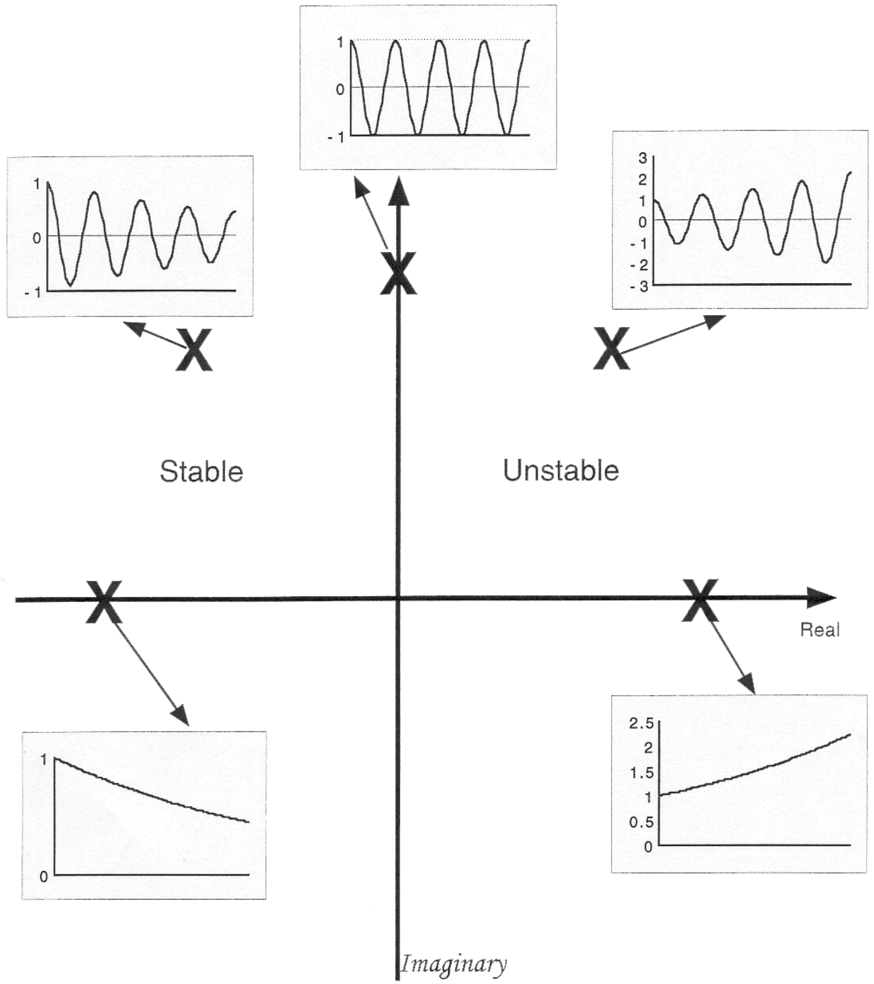


$$G(s)=\frac{Y(s)}{U(s)}=\frac{a_n s^n+a_{n-1}s^{n-1}+\ldots+a_0}{b_m s^m+b_{m-1}s^{m-1}+\ldots+b_0}$$

$$b_m s^m+b_{m-1}s^{m-1}+\ldots+b_0=0$$

Characteristic Equation

System Response



Single-track Model: Stability Analysis

$$\dot{X} = AX + B\delta$$

$$SX = AX(s) + B\delta(s)$$

$$(SI - A)X(s) = B\delta(s)$$

$$X(s) = (SI - A)^{-1}B\delta(s)$$

$$X(s) = \frac{(SI - A)^{*}B\delta(s)}{det(SI - A)}$$

$(SI - A)^{*}$. Adjoint Matric

Characteristic Equation

Single-track Model: Stability Analysis

- The characteristic equation of the bicycle handling model (see Slide 38) by substituting matrix A will be:

$$s^2 + a_1 s + a_0 = 0$$

$$s^2 + \underbrace{\frac{1}{u}\left[\frac{a^2 c_{\alpha f} + b^2 c_{\alpha r}}{I_z} + \frac{c_{\alpha f} + c_{\alpha r}}{m}\right]}_{a_1} s + \underbrace{\frac{1}{m I_z}\left[\frac{l^2 c_{\alpha f} c_{\alpha r}}{u^2} - m(a c_{\alpha f} - b c_{\alpha r})\right]}_{a_0} = 0, \quad l = a + b$$

<u>Routh's stability criterion</u>; Stability Condition: $if: a_0, a_1 \geq 0$

Single-track Model: Stability Conditions

$$\frac{1}{mI_z}\left[\frac{l^2 c_{\alpha_f} c_{\alpha_r}}{u^2} - m\left(ac_{\alpha_f} - bc_{\alpha_r}\right)\right] \geq 0$$

$$l + \underbrace{\left[\frac{-m(ac_{\alpha_f} - bc_{\alpha_r})}{lc_{\alpha_f} c_{\alpha_r}}\right]}_{k_{us}} u^2 \geq 0$$

Understeer Coefficient

Single-track Model: Stability Conditions

❖ Stability Condition:

$$if: \quad k_{us} > 0$$ → **Understeer Vehicle**

$$if: \quad k_{us} = 0$$ → **Neutral Steer Vehicle**

$$if: \quad k_{us} < 0$$ → **Oversteer Vehicle**

$$s^2 + a_1 s + a_0 = 0$$

$$s^2 + \underbrace{\frac{1}{u}\left[\frac{a^2 c_{\alpha f} + b^2 c_{\alpha r}}{I_z} + \frac{c_{\alpha f} + c_{\alpha r}}{m}\right]}_{a_1} S + \underbrace{\frac{1}{m I_z}\left[\frac{l^2 c_{\alpha f} c_{\alpha r}}{u^2} - m(a c_{\alpha f} - b c_{\alpha r})\right]}_{a_0} = 0$$

Stability Condition: $if: a_0, a_1 \geq 0$

$$l + k_{us} u^2 > 0$$

Understeer Vehicle

$$if: \quad k_{us} > 0$$

Neutral Steer Vehicle

$$if: \quad k_{us} = 0$$

$$\left.\begin{array}{l} \\ \end{array}\right\} \quad l + k_{us} u^2 > 0 \longrightarrow$$ **The Vehicle is always Stable**

Single-track Model: Stability Conditions

- **Stability Condition:**

Oversteer Vehicle

$$if: \quad k_{us} < 0 \longrightarrow l + k_{us} u^2 > 0 \longrightarrow$$

The equation has a root, known as critical speed

$$u_{cr} = \sqrt{\frac{-l}{k_{us}}}$$

$$if: \begin{cases} u < u_{cr} \longrightarrow \text{Stable Vehicle} \\ u \geq u_{cr} \longrightarrow \text{Unstable Vehicle} \end{cases}$$

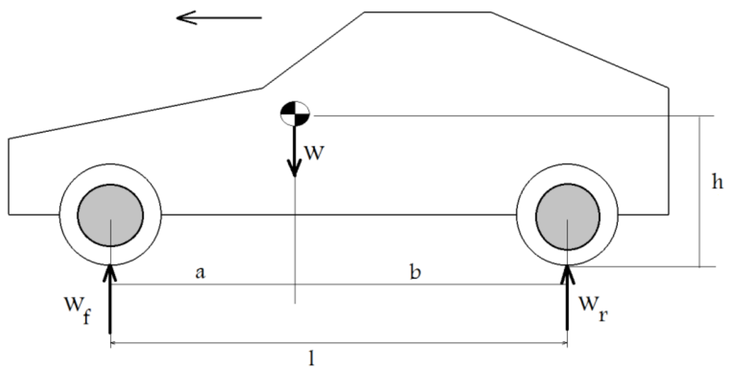
Note: An oversteer vehicle is stable for speeds less than the critical speed.

- **Stability Condition:**

$$k_{us} = \frac{-m\left(ac_{\alpha_f} - bc_{\alpha_r}\right)}{lc_{\alpha_f}c_{\alpha_r}}$$

$$k_{us} = \frac{bm}{l}\frac{1}{c_{\alpha_f}} - \frac{am}{l}\frac{1}{c_{\alpha_r}}$$

$$k_{us} = \frac{1}{g}(\underbrace{\frac{bmg}{l}}_{W_f}\frac{1}{c_{\alpha_f}} - \underbrace{\frac{amg}{l}}_{W_r}\frac{1}{c_{\alpha_r}})$$



$$\begin{cases} w_f = \frac{b}{l}\text{mg} \\ w_r = \frac{a}{l}\text{mg} \end{cases}$$

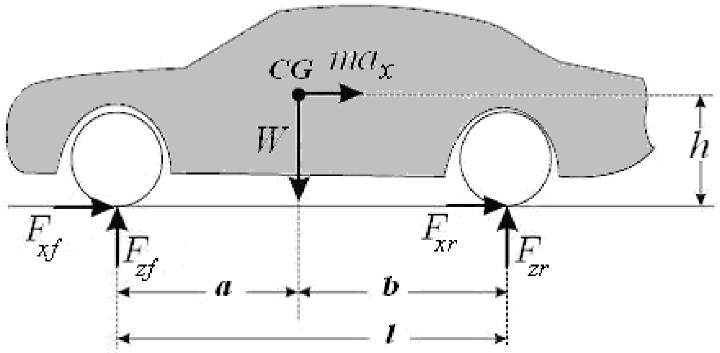
Alternative Form of Understeer Coefficient

$$k_{us} = \frac{1}{g}\left(\frac{w_f}{c_{\alpha_f}} - \frac{w_r}{c_{\alpha_r}}\right) \quad \text{deg}/\,g$$

Single-track Model: Understeer and Oversteer

- The relationship between understeer/oversteer characteristic and vehicle specifications:

$$k_{us} = \frac{m}{l c_{\alpha_f} c_{\alpha_r}} (b c_{\alpha_r} - a c_{\alpha_f})$$



Understeer Condition:

$$k_{us} > 0 \longrightarrow bc_{\alpha_r} - ac_{\alpha_f} > 0 \longrightarrow \left.\begin{array}{r} bc_{\alpha_r} > ac_{\alpha_f} \\ if\ C_{\alpha f} = C_{\alpha r} \end{array}\right] \longrightarrow b > a$$

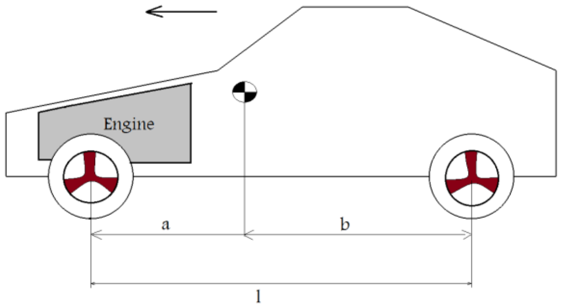
Oversteer Condition:

$$k_{us} < 0 \longrightarrow bc_{\alpha_r} - ac_{\alpha_f} < 0 \longrightarrow \left.\begin{array}{r} bc_{\alpha_r} < ac_{\alpha_f} \\ if\ C_{\alpha f} = C_{\alpha r} \end{array}\right] \longrightarrow b < a$$

Single-track Model: Understeer and Oversteer

❖ Typical Handling Characteristic of Some Types of Vehicles

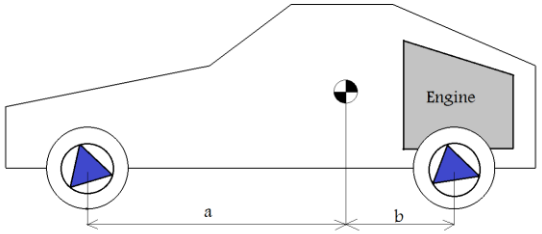
Front Engine/FWD



$b > a \quad C_{\alpha f} = C_{\alpha r}$

Understeer

Rear Engine/RWD

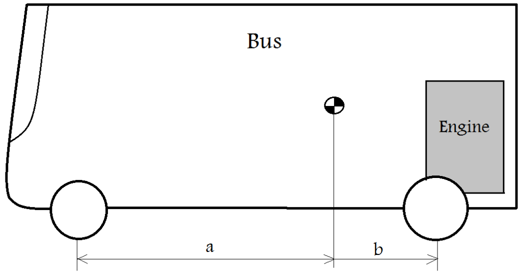


$b < a \quad C_{\alpha f} < C_{\alpha r}$

Neutral steer

Weak Understeer

Bus



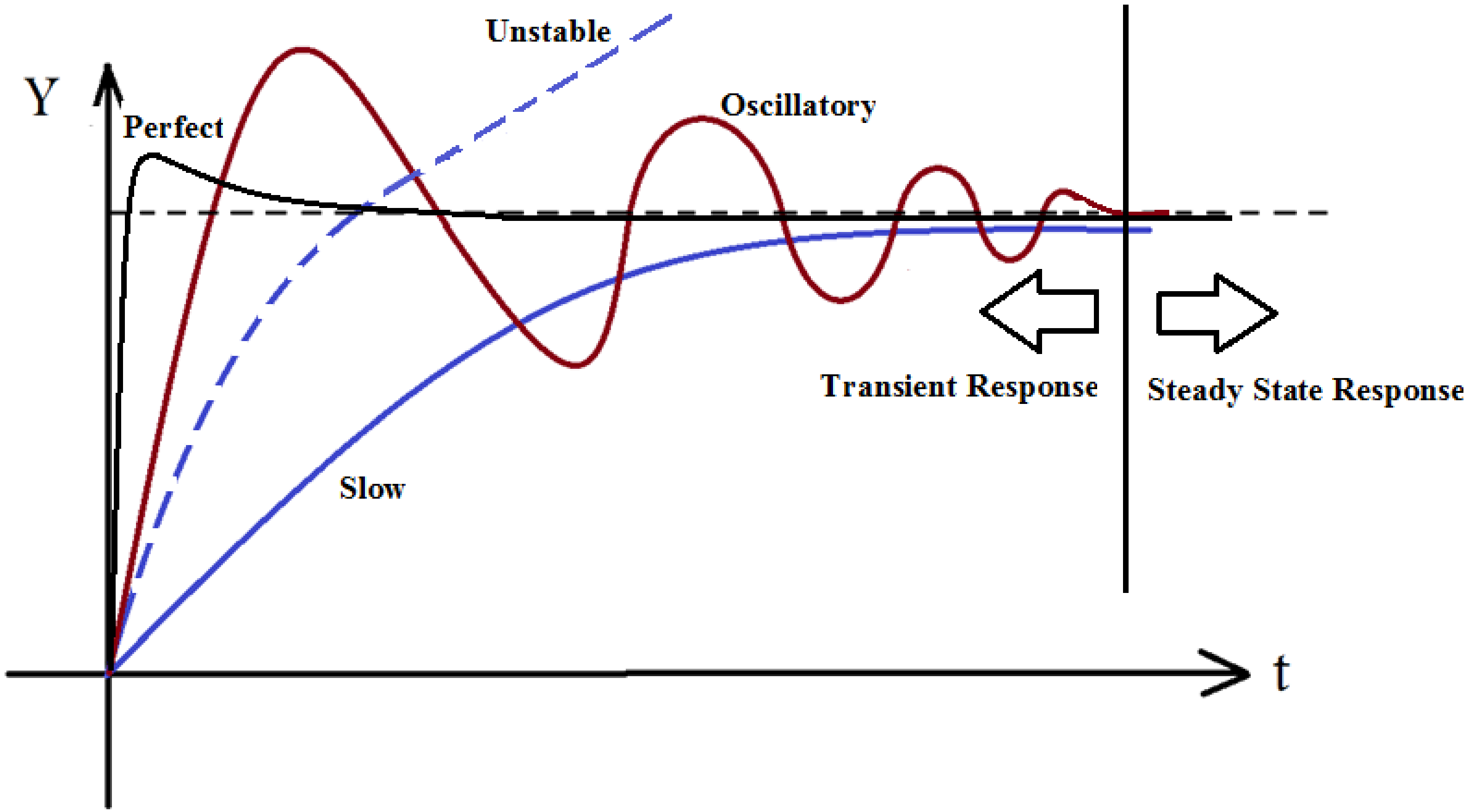
$$k_{us} = \frac{m}{l c_{\alpha_f} c_{\alpha_r}} (b c_{\alpha_r} - a c_{\alpha_f})$$

$b < a \quad C_{\alpha f} << C_{\alpha r}$

Understeer

Single-track Model: Transient and Steady State Response

- **Background**



Single-track Model: Handling Analysis

❖ Steady State Response Analysis of Bicycle Handling Model

$$\dot{X} = AX + B\delta$$

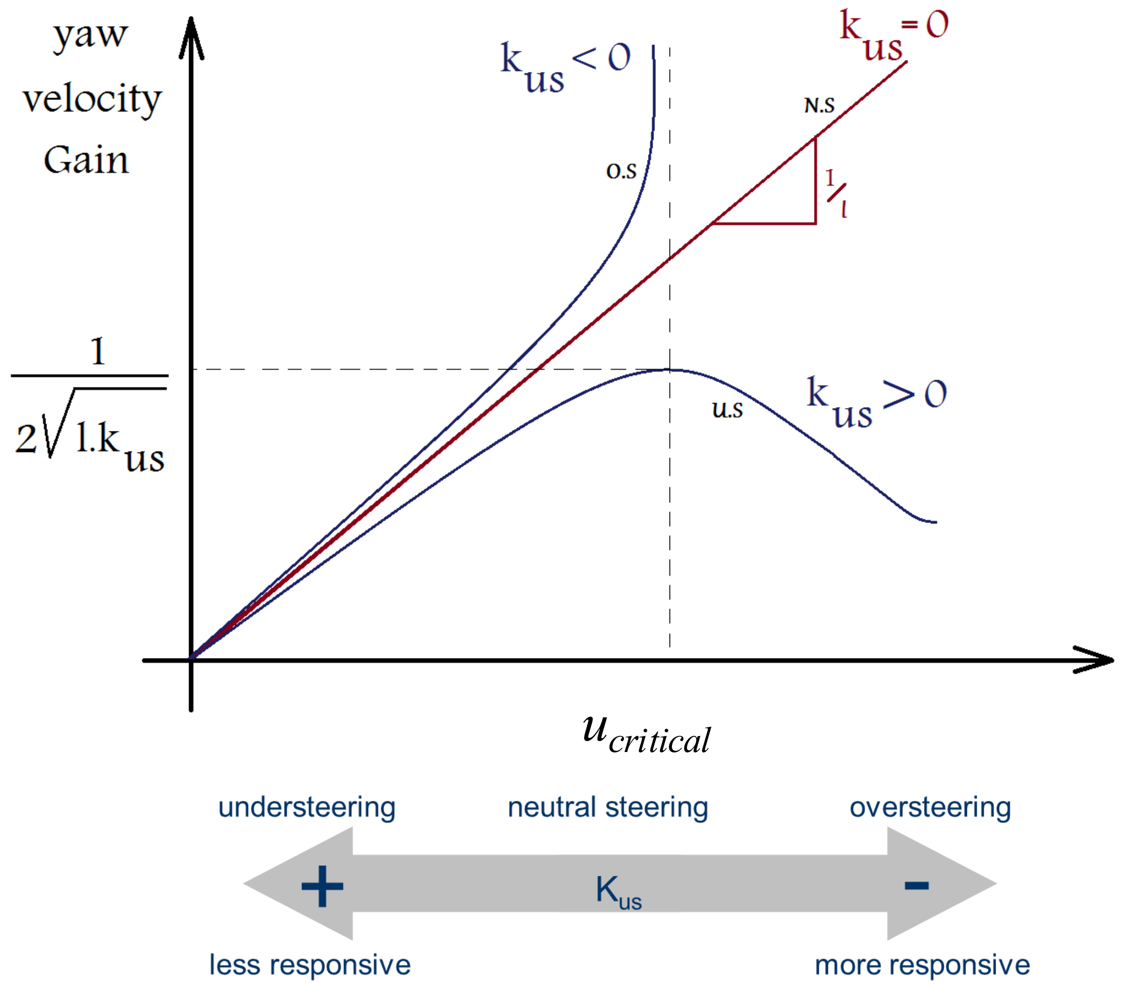
Steady State Condition $\quad \dot{X} = 0 \longrightarrow AX_{SS} + B\delta = 0$

$$\begin{bmatrix} r_{ss} \\ v_{ss} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^{-1} \begin{bmatrix} -b_1 \\ -b_2 \end{bmatrix} \delta$$

Yaw Velocity Gain	$\frac{r_{ss}}{\delta} = \frac{u}{l + k_{us} u^2}$
Sideslip Angle Gain	$\frac{\beta_{ss}}{\delta} = \frac{-(b + zu^2)}{l + k_{us} u^2} \quad \beta_{ss} = -\frac{v_{ss}}{u} \quad z = \frac{ma}{lc_{\alpha_r}}$
Lateral Acceleration Gain	$\frac{a_{y_{ss}}}{\delta} = \frac{u^2}{l + k_{us} u^2} \quad a_y = \dot{v} + ur \;,\; a_{y_{ss}} = ur_{ss}$

Single-track Model: Handling Analysis

❖ Steady State Response Analysis of Bicycle Handling Model



Similar Responses In Low Speeds

Yaw Velocity Gain

$$\frac{r_{ss}}{\delta} = \frac{u}{l + k_{us} u^2}$$

Thank you!